# Erisma — User manual

Version 0.4.0

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## 1. What Erisma is

Erisma is a suite of structural and geotechnical calculations to the Eurocodes. Every calculation is written out step by step — **input → formula → value → clause → check** — so that the engineer can verify each number by hand and submit the calculation sheet.

- **Results while you type:** checks, report and 3D model update after every change.
- **Worked examples as defaults:** every calculation opens with realistic values that pass, so you always start from a sensible design and change only what differs.
- **Hints:** every failing check says what to change (“increase the base width...”).

- **National Annexes:** partial factors and nationally determined parameters are read from the annex chosen in the project details and listed in every report.
- **Verified:** each module is tested against hand calculations and published values.

> Erisma is a tool for qualified engineers. The engineer remains responsible for the model, the input values and the interpretation of the results.

## 2. Installation and start

**Windows — installer (recommended)**

1. Run `Erisma-Setup-*version*.exe`. Python or anything else is not needed. Choose the language (Greek / English) and, if you wish, a desktop icon. No administrator rights are required (it installs for the current user; an administrator can choose all users).
2. Start **Erisma** from the Start menu or the desktop. A small window "Erisma — στατικοί υπολογισμοί" appears and the program opens in your browser. The window shows the address (normally `http://127.0.0.1:8000`; if that port is busy the next free one is used). **Άνοιγμα** opens the program again in the browser, **Τερματισμός** or closing the window stops Erisma.
3. The Start menu also has **Εγχειρίδιο Erisma** (this manual as PDF) and the uninstaller. Updating = running the new Setup; your licence and settings are kept.

**Without installer (Windows, Mac, Linux)**

1. Install Python 3.10 or newer (on Windows tick *Add Python to PATH*).
2. Unzip `Erisma.zip` into a folder of your choice.
3. **Windows:** double-click `Erisma.bat`. **Mac:** right-click `Erisma.command` → Open.

The first start installs the program (1–2 minutes, internet needed once); then Erisma opens in your browser at `http://127.0.0.1:8000`. To stop it close the black window.

Either way everything runs on your own computer and works offline; nothing is sent to the internet. Your projects are ordinary `.erisma` files that you keep wherever you like.

## 3. The screen

- **Home page:** all calculations as cards grouped by category, with search. "Continue" reopens the last calculation.
- **Left sidebar:** search (`Ctrl`+`K`), ★ favourites, recent calculations and all categories. « collapses it to icons.
- **Project details** (top of the form): company, project, job no., element, engineer, checker, date, **National Annex** and **project site**. They appear in the title block of every report and are kept for all calculations.
- **Input form** (middle): fields grouped in boxes. Above it the **status banner**: green when all checks pass, red with the failing checks and what to change.
- **Report / 3D model** (right): the full calculation sheet and the interactive model.

- **Above the form: + Add to project** (see “Projects”), 🎯 automatic design (most economical solution) and **Reset to defaults**. In the status banner the 💶 cost estimate.
- **Top bar:** Home, light/dark theme, language ΕΛ/EN, licence state, Manual, Open, Save, **DXF** (only for calculations with a drawing), Export HTML, Print / PDF.
- At the top of the left sidebar the 📁 **Project** list with the buttons 📄 project report, ⚖ comparison and ✚ new project.

## 4. Entering data

- **Decimals:** type `1,5` or `1.5` — both work.
- **Arithmetic:** type `2*3,5`, `(4+2)/2` or `1,5^2`; the result is shown next to the field and replaces the expression when you leave it.
- **Units** are shown next to every field; type the value in that unit.
- **Keyboard:** `Enter` goes to the next field, `Shift`+`Enter` back; `↑`/`↓` change the value in small steps (with `Shift` ×10).
- **Help:** the help text of a field appears while the field is selected, or click ⓘ.
- **Changed fields** are marked with a dot; ↺ returns that field to the example value. Each group shows how many of its fields you changed. “Reset to defaults” restores the whole example.
- **Groups** can be collapsed by clicking their title — Erisma remembers it.
- **Conditional fields** appear only when needed (e.g. jacket data only when “RC jacket” is chosen).
- **Tables** (soil layers, anchors, loads…): “+ Add row” copies the last row, ⧉ duplicates a row, × deletes it. You can **paste a block of cells from Excel**: select the first cell and press `Ctrl`+`V`.
- **Undo / redo:** `Ctrl`+`Z` / `Ctrl`+`Y`.
- **Wrong values** turn red immediately; if the calculation cannot run, the banner says why and the field concerned is highlighted.

## 5. Saving, files and printing

- **Autosave:** the inputs of every calculation are kept in the browser; when you reopen a calculation you continue where you left off.
- **Save** (`Ctrl`+`S`) downloads a file `.erisma` with the inputs and the project details — the whole project (all items, links, logo, unit prices) when the project list is not empty, otherwise the current calculation. **Open** (`Ctrl`+`O`) loads either kind again — send it to a colleague or keep it with the job.
- **Print / PDF** (`Ctrl`+`P`) prints the report; choose “Save as PDF” as the printer to get a PDF. **Export HTML** saves the report as a web page.

## 6. National Annex and project site

Choose the **National Annex** in the project details. All partial factors and nationally determined parameters (NDPs) come from it, and each report ends with a table of the NDPs used. Where Erisma has

no annex for a part (for example EN 1995-1-1 for Cyprus) the EN recommended values are used and the report says so.

With the Cyprus annex, 📍 **Map** opens the official seismic zonation map: click on the site, search a town, use your location or type coordinates. The seismic zone and $a_{gR}$ are filled in automatically (near a boundary the higher zone within 1 km is used), and the wind and snow calculation reads $v_{b,0}$ and $s_k$ for the same site. Next to every $a_{gR}$ field the zone buttons set the value with one click.

## 7. Reading the report

- The **title block** carries the project details; the summary lists every check with its utilisation.
- Each line shows **symbol = expression = value unit** with the clause of the standard.
- Checks show demand ≤ capacity, the utilisation U and **PASS / FAIL**; failing checks carry a hint in the status banner.
- Figures (sections, pressure diagrams, spectra) are drawn to scale from the actual values.
- The last table lists the National Annex parameters used.

## 8. The 3D model

Every calculation has an interactive 3D model (tab “3D model” or `Ctrl`+`3`). Drag to rotate, right-drag to pan, scroll to zoom. The **Layers** panel switches soil, loads, pressures, reinforcement, deflected shapes and labels on and off; the buttons give 3D, front and plan views and **PNG** saves an image for your report. The panel on the right lists the utilisation of every check; the model updates with every change.

## 9. Projects: several calculations together

A **project** collects all the calculations of one job (e.g. column C1, footing F1, retaining wall W1) so that they are saved, linked, priced and printed together.

1. Open a calculation and set its inputs.
2. Press **+ Add to project** (above the form) and give the element a name, e.g. “Footing F1”. The calculation is copied into the project; from now on the inputs you change are kept in that project item.
3. Repeat for every element. The same calculation can be added several times (F1, F2, ...).

The 📁 **Project** section at the top of the left sidebar lists the items. The dot in front of each name is green when all checks pass, red when a check fails and grey before the first run; hover over it to see the maximum utilisation U. 🔗*n* shows how many inputs are linked. Next to each item: ↑ moves it up (this is also the order in the project report), ✎ renames, × removes it (links that used it are removed too). The header buttons are 📄 project report, ⚖ comparison and ✚ new, empty project.

The project is autosaved in the browser. **Save** (`Ctrl`+`S`) then writes *one* `.erisma` file with all items, their inputs, links and the project details (file name from the job no. or project name); **Open** restores the whole project. **Reset to defaults** on a project item also removes its links.

## 10. Linking calculations (calculation chain)

An input of one project item can take its value from a **result** of another item, so a change travels down the chain by itself — e.g. the axial force of the column becomes the load on the footing.

1. Open the receiving item (e.g. “Footing F1”) from the project list. Every numeric input now has a 🔗 button next to its label.
2. Click 🔗. Choose the **source calculation**; the list shows only its results whose unit is of the **same kind** as the input (force with force, moment with moment, …), with symbol, description and current value. Units are converted automatically (e.g. kN → MN).
3. Optionally type a **factor** — e.g. `1/1,35` to turn a design value $N_{Ed}$ into a characteristic permanent load G — and tick **Use absolute value** for results that may be negative (e.g. hogging moments).
4. Press **Link**. The field becomes read-only and shows “🔗 from Column C1: N_Ed × 0,741”; **×** in that line removes the link.

Value used = factor × (|source result| if “absolute” is ticked, else source result). Before each run Erisma recalculates the whole project in dependency order (sources first), so the value is always current. If the source item is deleted or its result no longer exists, the field is marked red (“Link broken”) and the last value is kept. Links that form a loop (A → B → A) are refused with the message “Circular link between calculations” naming the items. Linked values are visible in the report as ordinary inputs, so the sheet remains checkable by hand.

## 11. Project report and company logo

📄 in the project header produces the whole project as one printable document:

- **Cover page** with your company logo, project name and the project details (company, job no., engineer, checker, date, National Annex).
- **Contents**: every item with its calculation, result (PASS / FAIL) and maximum utilisation.
- Every calculation sheet in full, in the order of the project list.

It opens in a new tab; print it with `Ctrl`+`P` or save it as PDF. **Logo:** in the project details press 🖼 **…** next to “Company logo” and choose an image (PNG, JPG, SVG…). It is reduced to at most 360 px to keep project files small, stored in the project file and printed on the cover and in the title block. “Remove logo” deletes it.

## 12. Quantities and cost estimate

Calculations of members that are built (footings, walls, beams, columns, slabs, piles, steel and timber members, strengthening…) end with a table **Quantities and cost estimate**: concrete, reinforcement, structural steel, timber, masonry, gabions, formwork, excavation, pile boring, ground anchors or FRP, each with quantity, unit price and cost. The total also appears in the status banner (💶 **≈ … €**; hover for the quantities). Section “Calculation reference” states for each calculation whether it has quantities.

- Quantities come from the actual geometry and the reinforcement chosen in the calculation (steel at 7850 kg/m³). Where the bar layout is not modelled in detail (e.g. walls, tanks) a typical reinforcement ratio is used and stated under the table.
- **Unit prices**: press 💶 **Unit prices** in the project details or ✎ on the cost chip. Defaults (€, supply and placing): concrete 130/m³, reinforcement 1,30/kg, structural steel 3,20/kg, timber 950/m³, masonry 160/m³, gabions 95/m³, formwork 25/m², excavation 12/m³, pile boring 70/m, anchors 90/m, FRP 120/m² per layer. Your prices are kept with the project and the project file; “Reset prices” returns to the defaults.

> The estimate is for comparing alternatives and preliminary budgets; it is not a bill of quantities (no laps, wastage, site overheads or VAT).

## 13. Automatic design — most economical solution

🎯 (above the form) searches for the **cheapest design that passes every check**. It is available for the calculations listed with “Automatic design” in the calculation reference, together with the variables it may change.

1. The dialog lists the design variables (e.g. footing $B_x$, $B_y$, h; beam depth, number and diameter of bars; steel section). Untick the ones that must stay as they are (e.g. a width fixed by the architect) and adjust **From / To / Step** (in the field's unit). Choice variables (sections, bar diameters) try all options.
2. Press **Search**. Every candidate is calculated in full with your other inputs, loads and National Annex. Ranking: first designs where all checks pass, then lowest cost (from the quantities and your unit prices), then lowest utilisation.
3. The result shows current and optimum values side by side (changed rows highlighted), cost and utilisation of both, number of trials, method and time. **Apply** puts the optimum into the form (`Ctrl`+`Z` undoes it).

**Method:** if the combinations are at most 2500 every one is tried (*exhaustive*, the true optimum on the grid). Otherwise *coordinate descent*: starting from your design, each variable in turn is run over its whole range and the best value kept, repeated (up to 5 rounds) until nothing improves — this gives a very good but not always the global optimum, so it is worth starting from a sensible design. A search stops after 40 s and returns the best found. If nothing passes in the ranges: “No passing design found — widen the ranges”. The optimum is an ordinary design: check it and round it to practical dimensions if necessary.

## 14. Comparing alternatives

⚖ in the project header shows all project items in one table: element, calculation, result (PASS / FAIL), maximum utilisation U, quantities and cost, with the project total Σ. To compare alternatives add the same element several times with different designs — e.g. “Wall – cantilever RC”, “Wall – gravity”, “Wall – gabions”, or two footing sizes. Among items of the **same calculation type** the cheapest one that passes is marked ★.

## 15. DXF drawings for CAD

For the calculations marked “DXF drawing” in the calculation reference the **DXF** button in the top bar downloads a drawing of the current design: plan and/or section with outline, reinforcement (bars and diameters, e.g. 5Ø16/200), soil line, dimensions and a title with element and project. It opens in AutoCAD and compatible CAD programs (BricsCAD, ZWCAD, progeCAD…) (format DXF R12, Greek code page, so Greek text is correct).

- **Units:** drawing units are millimetres at true size (1 unit = 1 mm); scale it at plotting.
- **Layers** (with colours): CONCRETE (7), REBAR (1 red), SOIL (34), DIM (3 green), TEXT (2 yellow), AXIS (8), STEEL (5 blue), HATCH (9) — so you can freeze or re-colour them to your office standard.
- The drawing is a starting point for the construction drawing; detailing (laps, anchorage lengths, bar schedules) is the designer's responsibility.

## 16. Trial and licence

Erisma can be used free of charge with all functions for **30 days** from the first start on a computer. The button in the top bar shows the state: “⏳ Trial: *n* days left”, “✓ Licensed to *name*” or “⚠ Trial ended”.

- **Activating:** click that button, paste the licence key you received (`ERISMA-…`; spaces and line breaks are ignored) and press **Activate**. The key is checked on your computer (no internet needed) and kept in your user profile (`%APPDATA%\Erisma` on Windows, `~/.config/erisma` on Mac/Linux), so it survives updates of the program.
- **After the trial** without a key, calculations, reports, project reports, comparison, automatic design and DXF are blocked and the licence window opens; your inputs and project files are kept and everything works again as soon as a key is entered. The manual stays available.
- A key carries the licensee's name, the plan and possibly an expiry date. A key that is not accepted gets a clear reply (“not an Erisma licence key”, “damaged — copy it again”, “not valid”, “expired on …”). When a key with an expiry date runs out the program falls back to the trial state. Keys are digitally signed (Ed25519) and cannot be altered or made up.

# 17. Keyboard shortcuts

| Keys | Action |
|---|---|
| `Ctrl+K` | Search calculations |
| `Ctrl+S` | Save project file |
| `Ctrl+O` | Open project file |
| `Ctrl+P` | Print / PDF |
| `Ctrl+3` | Report ↔ 3D model |
| `Ctrl+Z / Ctrl+Y` | Undo / redo inputs |
| `Enter / Shift+Enter` | Next / previous field |
| `↑ / ↓ (Shift)` | Change the value (×10) |
| `F1` | This manual |

# 18. National Annexes in Erisma

### EN recommended values

EN 1990/1992/1997/1998 recommended values

### Cyprus (CYS National Annexes)

NA to CYS EN 1990:2002, EN 1992-1-1:2004, EN 1997-1:2004, EN 1998-1:2004, EN 1998-5:2004 (approved 11/06/2010)

- Design Approach 2 is mandatory for STR/GEO (NA 2.9).
- Seismic zones: Zone 1 0.15 g, Zone 2 0.20 g, Zone 3 0.25 g (NA zonation map, 10 % in 50 years).

# 19. Calculation reference

ANALYSIS

## Continuous beam analysis

Standard: Stiffness method

Linear elastic analysis of single or multi-span beams with any number of supports, point loads, UDLs and moments. Gives reactions, shear, moment and deflection diagrams.

**Available tools:** 3D model

**Results that can be linked:** L, E, I, M_max,sag, M_max,hog, V_max, δ_max

### Method and assumptions

- Linear elastic stiffness (finite element) method, Euler–Bernoulli beam.
- Supports pinned or fixed at any position; loads downward positive.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Total length | m | 10 | 0.1 – … | |
| Material | | Steel (E = 210 GPa) | Steel (E = 210 GPa) / Concrete C25/30 (E = 31 GPa) / Timber C24 (E = 11 GPa) / Aluminium (E = 70 GPa) | |
| Second moment of area I | cm⁴ | 8356 | 0.001 – … | |
| Supports | table | | | Position [m], Type |
| Distributed loads | table | | | Start [m], End [m], w (down +) [kN/m] |
| Point loads | table | | | Position [m], P (down +) [kN] |
| Applied moments | table | | | Position [m], M (clockwise +) [kNm] |

CONCRETE

## Pile cap (strut-and-tie)

Standard: EN 1992-1-1 §6.5, §9.8.1

Rigid pile cap on 2, 3 or 4 piles under a column: pile loads with moments, strut-and-tie model, tie reinforcement, strut angle and node stresses at the column and the piles. Interactive 3D model.

**Available tools:** 3D model · quantities and cost

**Results that can be linked:** d, P_max, θ, T_Ed, A_s, σ_c,col, σ_c,pile

### Method and assumptions

- Rigid cap: pile loads from N and M; strut-and-tie with z = 0.9d, ties along the pile lines; node stresses by Eq. 6.60/6.61.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Number of piles | | 4 | 2 / 3 / 4 | |
| Pile diameter | mm | 600 | 250 – 1500 | |
| Pile spacing (centre to centre) | mm | 1800 | 600 – … | Usually ≥ 3 D. |
| Cap depth h | mm | 1000 | 400 – 3000 | |
| Column size (square) | mm | 500 | 200 – … | |
| Edge distance beyond the piles | mm | 300 | 150 – … | |
| Design axial load N_Ed | kN | 3500 | 0 – … | |
| Design moment M_x,Ed | kNm | 150 | | |
| Design moment M_y,Ed | kNm | 0 | | |
| Concrete class | | C30/37 | C25/30 / C30/37 / C35/45 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Cover to the tie bars | mm | 75 | 50 – … | |
| Tie bars per tie | | 7 | 2 – 30 | |
| Tie bar Ø | | 20 | 16 / 20 / 25 / 32 | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| Strut angle θ ≥ 26.5° (cot θ ≤ 2) | 6.5.2 / 6.2.3 | 0.72 |
| Tie: T_Ed ≤ A_s f_yd | 6.5.3 | 0.91 |
| Column node (C-C-C): σ ≤ 1.0 ν' f_cd | Eq. 6.60 | 0.8 |
| Pile node (C-C-T): σ ≤ 0.85 ν' f_cd | Eq. 6.61 | 0.22 |

| | | |
|---|---|---|
| Edge distance beyond the pile face ≥ 150 mm | good practice | 0.5 |

## RC beam section design

Standard: EN 1992-1-1:2004

Rectangular section with tension and compression reinforcement: flexure (stress block, strain compatibility), minimum/maximum reinforcement and shear with variable strut inclination.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Overall depth h, Number of tension bars, Tension bar diameter)

**Results that can be linked:** b, h, c_nom, d, f_ck, f_cd, f_ctm, f_yk, f_yd, M_Ed, μ, μ_lim, ξ, z …

### Method and assumptions

- Rectangular stress block (λ = 0.8, η = 1.0), ε_cu3 = 3.5 ‰, x/d ≤ 0.45 for ductility.
- Shear with variable strut inclination 1 ≤ cot θ ≤ 2.5 (6.2.3).

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Width b | mm | 300 | 100 – … | |
| Overall depth h | mm | 500 | 150 – … | |
| Nominal cover to links | mm | 30 | 10 – … | |
| Concrete class | | C25/30 | 7 … | |
| Reinforcement fyk | N/mm² | 500 | 200 – 600 | |
| γc | | 1.5 | 1 – … | |
| γs | | 1.15 | 1 – … | |
| αcc | | 1 | 0.8 – 1 | 1.0 recommended (Greek NA); 0.85 UK NA. |
| Design moment MEd | kNm | 140 | 0 – … | |
| Design shear VEd | kN | 120 | 0 – … | |
| Number of tension bars | | 4 | 2 – … | |
| Tension bar diameter | | 16 | 8 … | |
| Number of compression (top) bars | | 2 | 2 – … | |
| Compression bar diameter | | 12 | 8 … | |
| Link diameter | | 8 | 8 / 10 / 12 / 14 | |
| Link legs | | 2 | 2 – … | |
| Link spacing | mm | 200 | 50 – … | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| Tension reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.95 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.2.1.1(1) | 0.23 |
| Maximum reinforcement: A_s + A_s2 ≤ A_s,max | 9.2.1.1(3) | 0.17 |
| Bar spacing: s_clear ≥ max(φ, d_g + 5, 20) | 8.2(2) | 0.47 |
| Moment resistance: M_Ed ≤ M_Rd | 6.1 | 0.96 |
| Concrete strut: V_Ed ≤ V_Rd,max | 6.2.3(3) | 0.32 |
| Shear reinforcement: V_Ed ≤ V_Rd,s | 6.2.3(3) | 0.54 |
| Minimum links: A_sw/(s b) ≥ ρ_w,min | 9.2.2(5) | 0.48 |
| Link spacing: s ≤ 0.75d | 9.2.2(6) | 0.59 |

## RC column

Standard: EN 1992-1-1 §5.8, EN 1998-1 §5.4

Rectangular column: imperfections, slenderness, second-order effects (nominal curvature), biaxial bending (5.8.9), shear, detailing and EC8 DCM rules. Interactive 3D with reinforcement.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Width b, Depth h (bending about the strong axis), Longitudinal bars Ø)

**Results that can be linked:** b × h, f_cd, f_yd, A_s, N_Ed, ω, n, N_Rd, l_0,h, λ_h, λ_lim,h, e_i,h, M_Ed,h, M_Rd,h …

### Method and assumptions

- Second-order effects by the nominal curvature method (5.8.8); biaxial by Eq. 5.39.
- M_Rd by strain compatibility for the actual axial force; EC8 DCM rules optional.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Width b | mm | 400 | 200 – … | |
| Depth h (bending about the strong axis) | mm | 400 | 200 – … | |
| Cover to links | mm | 30 | 15 – … | |
| Concrete class | | C25/30 | 7 … | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Longitudinal bars Ø | | 16 | 9 … | |
| Bars along b (each face, incl. corners) | | 4 | 2 – … | |

| | | | | |
|---|---|---|---|---|
| Bars along h (each face, incl. corners) | | 4 | 2 – … | |
| Links Ø | | 8 | 8 / 10 / 12 | |
| Link spacing | mm | 100 | 50 – … | |
| Clear height | m | 3 | 0.5 – … | |
| Effective length factor (bending in h) | | 1 | 0.5 – 2.5 | 0.5 fixed–fixed, 0.7 fixed–pinned, 1.0 pinned–pinned, 2.0 cantilever (or EN 1992-1-1 Eq. 5.15/5.16). |
| Effective length factor (bending in b) | | 1 | 0.5 – 2.5 | |
| Braced column | | yes | | Unbraced columns use r_m = 1.0 (C = 0.7). |
| Axial force N_Ed (compression +) | kN | 1200 | 0 – … | |
| M_Ed (h) at top | kNm | 60 | | First-order end moments; same sign = single curvature. |
| M_Ed (h) at bottom | kNm | -30 | | |
| M_Ed (b) at top | kNm | 20 | | |
| M_Ed (b) at bottom | kNm | -10 | | |
| Shear V_Ed | kN | 60 | 0 – … | |
| Effective creep ratio φ_ef | | 0.8 | 0 – 3 | |
| Primary seismic column (EN 1998-1 DCM) | | yes | | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| Axial: N_Ed ≤ N_Rd | 6.1 | 0.32 |
| Bending in h: M_Ed ≤ M_Rd | 6.1 | 0.28 |
| Bending in b: M_Ed ≤ M_Rd | 6.1 | 0.12 |
| Biaxial: (M_Ed,h/M_Rd,h)^a + (M_Ed,b/M_Rd,b)^a ≤ 1 | EN 1992-1-1 Eq. 5.39 | 0.31 |
| Shear: V_Ed ≤ max(V_Rd,c, min(V_Rd,s, V_Rd,max)) | 6.2 | 0.17 |
| A_s ≥ A_s,min = max(0.10 N_Ed/f_yd, 0.002 A_c) | 9.5.2(2) | 0.13 |
| A_s ≤ 0.04 A_c | 9.5.2(3) | 0.38 |
| Link spacing ≤ min(20φ, b_min, 400) | 9.5.3(3) | 0.31 |
| Axial load ratio v_d ≤ 0.65 | EN 1998-1 5.4.3.2.1(3) | 0.69 |
| ρ_l ≥ 1 % | 5.4.3.2.2(1) | 0.66 |
| ρ_l ≤ 4 % | 5.4.3.2.2(1) | 0.38 |

| | | |
|---|---|---|
| Critical region links: s ≤ min(b_0/2, 175, 8d_bL) | 5.4.3.2.2(11) | 0.78 |

## RC shear wall (EC8 DCM)

Standard: EN 1998-1 §5.4, EN 1992-1-1

Ductile cantilever wall of ductility class DCM: geometry, normalised axial load, critical region, flexure with boundary elements, shear with the amplification ε = 1.5, web reinforcement and confinement of the boundary elements (Eq. 5.20). Interactive 3D model.

**Available tools:** 3D model · quantities and cost

**Results that can be linked:** h_cr, v_d, A_s,be, M_Rd, V_Ed, V_Rd,max, V_Rd,s, μ_φ, ω_wd, α

### Method and assumptions

- DCM wall: v_d ≤ 0.4, h_cr (Eq. 5.19), M_Rd by strain compatibility with web bars, V_Ed = 1.5 V'_Ed, confinement of boundary elements by Eq. 5.20 with μ_φ = 2q_0 − 1.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Wall length l_w | m | 4 | 1 – 12 | |
| Wall thickness b_w | mm | 250 | 150 – 600 | |
| Clear storey height h_s | m | 3 | 2 – 6 | |
| Total wall height H_w | m | 18 | 3 – 120 | |
| Number of storeys | | 6 | 1 – 40 | |
| Axial force N_Ed (seismic, compression +) | kN | 2500 | 0 – … | |
| Moment M_Ed | kNm | 9000 | 0 – … | |
| Shear V'_Ed (from the analysis) | kN | 700 | 0 – … | |
| Basic behaviour factor q_0 | | 3 | 1.5 – 4.5 | DCM wall systems: q_0 = 3.0 (Table 5.1). |
| Concrete class | | C30/37 | C20/25 / C25/30 / C30/37 / C35/45 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Cover to the hoops | mm | 30 | 20 – … | |
| Boundary element length l_c | mm | 700 | 200 – … | |
| Bars in each boundary element | | 8 | 4 – 30 | |
| Boundary bar Ø | | 20 | 10 / 12 / 14 / 16 / 20 / 25 | |
| Boundary hoop Ø | | 10 | 8 / 10 / 12 | |

| | | | | |
|---|---|---|---|---|
| Boundary hoop spacing | mm | 85 | 50 – 250 | |
| Web vertical bars Ø (each face) | | 10 | 10 / 12 / 14 / 16 / 20 / 25 | |
| Web vertical spacing | mm | 200 | 75 – 400 | |
| Web horizontal bars Ø (each face) | | 12 | 10 / 12 / 14 / 16 / 20 / 25 | |
| Web horizontal spacing | mm | 150 | 75 – 400 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Thickness b_w ≥ max(150 mm, h_s/20) | 5.4.1.2.3(1) | 0.6 |
| Normalised axial load v_d ≤ 0.4 | 5.4.3.4.1(2) | 0.31 |
| Flexure: M_Ed ≤ M_Rd | 5.4.3.4.1 | 0.98 |
| Boundary element ρ ≥ 0.5 % | 5.4.3.4.2(8) | 0.35 |
| Boundary length l_c ≥ max(0.15 l_w, 1.5 b_w) | 5.4.3.4.2(6) | 0.86 |
| Web crushing: V_Ed ≤ V_Rd,max | 6.2.3 | 0.25 |
| Shear: V_Ed ≤ V_Rd,s | 6.2.3 | 0.5 |
| Web vertical ρ_v ≥ 0.2 % | EN 1992-1-1 9.6.2 | 0.64 |
| Web horizontal ρ_h ≥ max(0.25 ρ_v, 0.1 %) | EN 1992-1-1 9.6.3 | 0.17 |
| Confinement: α ω_wd ≥ 30 μ_φ (v_d + ω_v) ε_sy,d b_c/b_0 − 0.035 | Eq. 5.20 | 0.34 |
| Boundary ω_wd ≥ 0.08 | 5.4.3.4.2(4) | 0.18 |
| Hoop spacing ≤ min(b_0/2, 175, 8 d_bL) | 5.4.3.2.2(11) | 0.94 |

## RC slab

Standard: EN 1992-1-1

One-way continuous slabs with pattern loading, or two-way rectangular panels with any edge condition (plate analysis); bottom/top reinforcement, shear, deflection (span/depth) and punching at a column.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Slab thickness h, Bottom bar spacing, Top bar spacing)

**Results that can be linked:** g_k, q_k, p_d, m_x,span, m_y,span, m_x,sup, m_y,sup, μ, μ_lim, ξ, z, A_s,req, A_s,min, A_s,max …

**Method and assumptions**

- Two-way panels: Kirchhoff plate by finite differences with any edge conditions (verified against Timoshenko's tables).

- One-way continuous slabs: pattern loading (all / odd / even spans).

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Slab type | | Two-way panel | Two-way panel / One-way continuous | |
| Short span l_x | m | 4.5 | 1 – … | |
| Long span l_y (two-way) | m | 6 | 1 – … | |
| Number of equal spans (one-way) | | 3 | 1 – 8 | |
| Edge x = 0 | | Fixed / continuous | Simply supported / Fixed / continuous | |
| Edge x = l_x | | Simply supported | Simply supported / Fixed / continuous | |
| Edge y = 0 | | Fixed / continuous | Simply supported / Fixed / continuous | |
| Edge y = l_y | | Simply supported | Simply supported / Fixed / continuous | |
| Slab thickness h | mm | 180 | 100 – … | |
| Cover | mm | 25 | 15 – … | |
| Concrete class | | C25/30 | C20/25 / C25/30 / C30/37 / C35/45 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Superimposed permanent load (excl. self weight) | kN/m² | 2 | 0 – … | Finishes, screed, partitions. |
| Imposed load | kN/m² | 2 | 0 – … | EN 1991-1-1: residential 1.5–2.0, offices 3.0, balconies 4.0. |
| Bottom bars Ø | | 10 | 8 / 10 / 12 / 14 / 16 | |
| Bottom bar spacing | mm | 150 | 75 – 400 | |
| Top bars Ø (supports) | | 12 | 8 / 10 / 12 / 14 / 16 | |
| Top bar spacing | mm | 150 | 75 – 400 | |
| Check punching at a column (flat slab) | | no | | |
| Column size c1 | mm | 400 | 150 – … | |
| Column size c2 | mm | 400 | 150 – … | |
| Column reaction V_Ed | kN | 450 | 0 – … | |
| Column position | | Internal | Internal / Edge / Corner | β = 1.15 / 1.4 / 1.5 (EN 1992-1-1 Fig. 6.21N). |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.35 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.3.1.1 | 0.38 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.34 |
| Bar spacing at max. moment: s ≤ min(2h, 250) | 9.3.1.1(3) | 0.6 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.23 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.3.1.1 | 0.36 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.23 |
| Bar spacing at max. moment: s ≤ min(2h, 250) | 9.3.1.1(3) | 0.6 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.49 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.3.1.1 | 0.26 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.5 |
| Bar spacing at max. moment: s ≤ min(2h, 250) | 9.3.1.1(3) | 0.6 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.4 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.3.1.1 | 0.26 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.41 |
| Bar spacing at max. moment: s ≤ min(2h, 250) | 9.3.1.1(3) | 0.6 |
| Shear: V_Ed ≤ V_Rd,c | 6.2.2 | 0.41 |
| Span/depth: l/d ≤ (l/d)_lim | 7.4.2 | 0.59 |

## RC stair flight

Standard: EN 1992-1-1, EN 1991-1-1

Straight flight with a waist slab spanning between landings or beams: geometry rules (2R + G), self weight of waist and steps, bending, shear, minimum reinforcement and span/depth. Interactive 3D model.

**Available tools:** 3D model · quantities and cost

**Results that can be linked:** R, G, α, g_sw, g_k, p_d, M_Ed, V_Ed, d, μ, μ_lim, ξ, z, A_s,req ...

**Method and assumptions**

- Waist slab as a one-way slab on the horizontal span; self weight of waist and steps on plan; comfort rule 2R + G.

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Horizontal span (support to support) | m | 3.08 | 1 – 8 | |
| Total rise of the flight | m | 2.04 | 0.3 – 4 | |
| Number of risers | | 12 | 2 – 25 | |
| Flight width | m | 1.2 | 0.6 – 4 | |
| Waist thickness (perpendicular to the slope) | mm | 200 | 100 – 350 | |
| Support condition | | Simply supported | Simply supported / Continuous with landings | Continuous: M = wL²/10 (end spans with landing slabs). |
| Finishes (on plan) | kN/m² | 1.5 | 0 – … | |
| Imposed load | kN/m² | 3 | 0 – … | EN 1991-1-1 category A stairs: 2.0–4.0 kN/m² (national annex). |
| Concrete class | | C25/30 | C20/25 / C25/30 / C30/37 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Cover | mm | 25 | 15 – … | |
| Main bars Ø | | 12 | 8 / 10 / 12 / 14 / 16 | |
| Main bar spacing | mm | 150 | 75 – 300 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Comfort: 2R + G ≤ 650 mm | building regulations (600–650) | 0.95 |
| Comfort: 2R + G ≥ 580 mm | building regulations (600–650) | 0.94 |
| Riser R ≤ 190 mm | building regulations | 0.89 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.39 |
| Minimum reinforcement | 9.3.1.1 | 0.3 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.4 |
| Bar spacing ≤ min(3t, 400) | 9.3.1.1(3) | 0.38 |
| Shear: V_Ed ≤ V_Rd,c | 6.2.2 | 0.25 |
| Span/depth: l/d ≤ (l/d)_lim (horizontal span) | 7.4.2 | 0.91 |

## Water tank / pool

Standard: EN 1992-3, EN 1992-1-1, EN 1997-1

Rectangular reinforced concrete tank or swimming pool: wall bending for the full (water test) and empty (backfill + groundwater) cases, crack width to EN 1992-3 tightness class 1, minimum thickness and reinforcement, and flotation (UPL) of the empty tank. Interactive 3D.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Wall thickness, Spacing, inside face, Spacing, outside face)

**Results that can be linked:** H, h_w, t, w_k1, K_0, M_Ed,base, M_qp,base, d, M_Ed, μ, μ_lim, ξ, z, A_s,req …

### Method and assumptions

- Long walls as vertical strips fixed into the base, free or pinned at the top.
- Crack width by EN 1992-1-1 7.3.4 (quasi-permanent) against w_k1 of tightness class 1.
- Thermal/early-age restraint cracking is not included.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Internal length | m | 10 | 1 – … | |
| Internal width | m | 5 | 1 – … | |
| Wall height (above base) | m | 2.5 | 0.5 – 12 | |
| Maximum water depth | m | 2.3 | 0.1 – … | |
| Wall thickness | mm | 250 | 120 – 1000 | |
| Base slab thickness | mm | 300 | 150 – 1500 | |
| Base projection beyond the walls | m | 0.3 | 0 – … | Soil resting on the projection helps against flotation. |
| Wall support at the top | | Free (open tank / pool) | Free (open tank / pool) / Pinned (roof slab) | |
| Backfill height outside (above base) | m | 2.5 | 0 – … | |
| Backfill γ | kN/m³ | 19 | 10 – … | |
| Backfill φ'k | ° | 30 | 20 – 45 | |
| Surcharge on the backfill | kN/m² | 5 | 0 – … | |
| Groundwater outside (above base underside) | m | 1 | 0 – … | Highest expected level — governs flotation of the empty tank. |
| Concrete class | | C30/37 | C30/37 / C35/45 / C40/50 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Cover to bars | mm | 40 | 25 – 75 | |

| | | | | |
|---|---|---|---|---|
| Vertical bars, inside face Ø | | 12 | 10 / 12 / 14 / 16 / 20 | |
| Spacing, inside face | mm | 125 | 75 – 300 | |
| Vertical bars, outside face Ø | | 12 | 10 / 12 / 14 / 16 / 20 | |
| Spacing, outside face | mm | 125 | 75 – 300 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Minimum wall thickness (EN 1992-3 9.11.1, classes 1–2) | EN 1992-3 9.11.1(102) | 0.6 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.31 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.31 |
| A_s ≥ A_s,min = k_c k f_ct,eff A_ct / f_yk (k_c = 0.4) | EN 1992-1-1 7.3.2 | 0.32 |
| Crack width inside: w_k ≤ w_k1 | EN 1992-3 7.3.1 | 0.56 |
| Compression zone x ≥ x_min = min(50 mm, 0.2t) (no through-cracks) | EN 1992-3 7.3.1(112) | 0.81 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.61 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.61 |
| A_s ≥ A_s,min = k_c k f_ct,eff A_ct / f_yk (k_c = 0.4) | EN 1992-1-1 7.3.2 | 0.32 |
| Crack width outside: w_k ≤ w_k1 | EN 1992-3 7.3.1 | 0.78 |
| Compression zone x ≥ x_min = min(50 mm, 0.2t) (no through-cracks) | EN 1992-3 7.3.1(112) | 0.81 |
| Shear: V_Ed ≤ V_Rd,c | 6.2.2 | 0.46 |
| UPL: γ_G,dst U ≤ γ_G,stb G (1 / 0.9) | EN 1997-1 Eq. 2.8 | 0.5 |

## EXISTING BUILDINGS

### Assessment and strengthening of RC members

Standard: EN 1998-3 Annex A

Existing RC column or beam: knowledge level and confidence factor, chord rotation at yield and ultimate (A.10a, A.1), cyclic shear resistance (A.12), checks for the limit states NC, SD and DL, and strengthening with an RC jacket or FRP wrapping. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · automatic design (Jacket thickness, Jacket hoop spacing)

**Results that can be linked:** CF, f_c, f_y, f_c,V, f_yw,V, φ_y, M_y, a_v, θ_y, θ_um, θ_SD, V_R, V_My, b* × h* ...

**Method and assumptions**

- EN 1998-3 Annex A: θ_y (A.10a), θ_um (A.1), V_R (A.12); mean strengths ÷ CF, brittle shear also ÷ γ (Cyprus NA 2.3).
- NC: θ_E ≤ θ_um, SD: θ_E ≤ ¾θ_um, DL: θ_E ≤ θ_y. RC jacket by the monolithic factors of A.4.2.2, FRP by A.4.4. The Greek KANEPE is not implemented.

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Knowledge level | | KL2 — normal knowledge | KL1 — limited knowledge / KL2 — normal knowledge / KL3 — full knowledge | KL1: simulated design + limited in-situ tests; KL2: extended inspection and testing; KL3: comprehensive (Table 3.1). |
| Primary seismic element | | yes | | γ_el = 1.5 (deformation) and 1.15 (shear) for primary elements, 1.0 for secondary. |
| Detailed for earthquake resistance | | no | | Off for older buildings (no seismic detailing): θ_um is divided by 1.2 (A.3.2.2). |
| Ribbed (deformed) bars | | yes | | |
| Width b | mm | 300 | 150 – … | |
| Depth h (in the direction of bending) | mm | 400 | 150 – … | |
| Cover to bar centre | mm | 35 | 15 – … | |
| Bars along b (each face, incl. corners) | | 3 | 2 – … | |
| Bars along h (each face, incl. corners) | | 3 | 2 – … | |
| Longitudinal bar Ø | mm | 16 | 8 – 32 | |
| Hoop Ø | mm | 8 | 6 – 14 | |
| Hoop spacing | mm | 200 | 50 – 400 | |
| Hoop legs parallel to the shear | | 2 | 2 – 6 | |
| Concrete mean strength f_cm (tests) | N/mm² | 16 | 8 – 60 | |

| | | | | |
|---|---|---|---|---|
| Steel mean yield strength f_ym | N/mm² | 450 | 220 – 700 | |
| Shear span L_V = M/V | m | 1.5 | 0.3 – … | Half the clear height for a column in double curvature. |
| Axial force in the seismic situation | kN | 600 | 0 – … | Compression positive (G + ψ2Q). |
| Chord rotation demand θ_E — NC | | 0.02 | 0 – 0.2 | rad, from the analysis for the NC seismic action. |
| Chord rotation demand θ_E — SD | | 0.012 | 0 – 0.2 | |
| Chord rotation demand θ_E — DL | | 0.004 | 0 – 0.2 | |
| Shear demand V_E | kN | 150 | 0 – … | From capacity design: ΣM_y / L_cl of the member, or from the analysis. |
| Strengthening | | RC jacket | None (assessment only) / RC jacket / FRP wrapping | |
| Jacket thickness | mm | 100 | 60 – 250 | ≥ 75 mm cast, ≥ 50 mm shotcrete. |
| Jacket concrete mean strength | N/mm² | 33 | 20 – 60 | |
| Jacket bars along each face (incl. corners) | | 3 | 2 – … | |
| Jacket bar Ø | mm | 16 | 10 – 32 | |
| Jacket hoop Ø | mm | 10 | 8 – 14 | |
| Jacket hoop spacing | mm | 100 | 50 – 200 | |
| Jacket steel mean yield strength | N/mm² | 550 | 400 – 700 | |
| FRP layers | | 2 | 1 – 10 | |
| Thickness per layer | mm | 0.17 | 0.05 – 2 | |
| FRP modulus E_f | GPa | 230 | 20 – 650 | |
| FRP ultimate strain ε_u,f | | 0.015 | 0.003 – 0.05 | |

| | | | | |
|---|---|---|---|---|
| Corner radius | mm | 25 | 10 – 60 | Round the corners before wrapping (≥ 20–25 mm). |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| NC (with RC jacket): θ_E ≤ θ_um | EN 1998-3 A.3.2.2 | 0.72 |
| SD (with RC jacket): θ_E ≤ ¾ θ_um | EN 1998-3 A.3.2.3 | 0.58 |
| DL (with RC jacket): θ_E ≤ θ_y | EN 1998-3 A.3.2.4 | 0.62 |
| Shear (with RC jacket): V_E ≤ V_R | EN 1998-3 A.3.3.1 | 0.44 |

FIRE

## Fire resistance — RC beam / slab

Standard: EN 1992-1-2 §5 (tabulated data)

Tabulated data for simply supported beams (Table 5.5) and slabs (Table 5.8): minimum width / thickness and axis distance of the reinforcement for the required fire resistance.

**Available tools:** 3D model

**Results that can be linked:** a, a_req, a_sd

**Method and assumptions**

- Tabulated data: Table 5.5 (simply supported beams, interpolated in b, corner bars +10 mm) and Table 5.8 (slabs).

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Member | | Simply supported beam | Simply supported beam / One-way slab / Two-way slab (l_y/l_x ≤ 1.5) / Two-way slab (1.5 < l_y/l_x ≤ 2) | |
| Required fire resistance | | R90 | R30 / R60 / R90 / R120 / R180 / R240 | |
| Beam width b / slab thickness h | mm | 300 | 60 – … | |
| Nominal cover to the links (beam) / bars (slab) | mm | 35 | 10 – … | |
| Link Ø (beam) | mm | 8 | 0 – 16 | |
| Main bar Ø | mm | 20 | 8 – 40 | |
| One layer of bars (beam: corner bars need a + 10 | | yes | | |

| | | | | |
|---|---|---|---|---|
| mm) | | | | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Beam width b ≥ b_min (R90) | EN 1992-1-2 Table 5.5 | 0.5 |
| Axis distance a ≥ a_req (R90) | EN 1992-1-2 5.6 | 0.94 |

## Fire resistance — steel member

Standard: EN 1993-1-2, EN 1991-1-2

Steel I/H member in the standard fire: degree of utilisation, critical temperature, section factor with shadow effect, heating of the unprotected or protected member and the fire resistance time against the requirement.

**Available tools:** 3D model

**Results that can be linked:** A_m/V, [A_m/V]_b, k_sh, μ_0, θ_a,cr, A_p/V, λ_p, d_p, t_fi,d

**Method and assumptions**

- Critical temperature Eq. 4.22; heating to ISO 834 by Eq. 4.25 (unprotected, k_sh) or Eq. 4.27 (protected, product data λ_p, d_p, c_p, ρ_p).

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Section | | HEB 240 | 40 … | |
| Fire exposure | | 3 sides (beam under a slab) | 4 sides (column) / 3 sides (beam under a slab) | |
| Utilisation at normal temperature E_d / R_d | | 0.8 | 0.05 – 1 | From the cold design (e.g. the steel beam calculation). |
| Reduction factor for fire η_fi | | 0.65 | 0.2 – 1 | E_fi,d = η_fi E_d; EN 1993-1-2 2.4.2(3): 0.65 recommended (0.7 for storage areas). |
| Required fire resistance | | R60 | R30 / R60 / R90 / R120 / R180 / R240 | |
| Fire protection | | Board / spray (λ_p, d_p) | None / Board / spray (λ_p, d_p) | |
| Thermal conductivity λ_p | | 0.12 | 0.02 – 1 | W/mK — from the product data. |

| | | | | |
|---|---|---|---|---|
| Protection thickness d_p | mm | 20 | 2 – 100 | |
| Specific heat c_p | | 1200 | 100 – 3000 | J/kgK |
| Density ρ_p | kN/m³ | 3 | 0.5 – 20 | e.g. 3 kN/m³ ≈ 300 kg/m³ |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Fire resistance ≥ R60 | EN 1993-1-2 2.4.2 | 0.36 |

## Fire resistance — timber beam

Standard: EN 1995-1-2 §4.2.2

Reduced cross-section method: charring depth, effective section, strength in fire (k_fi, k_mod,fi = 1) and bending, lateral buckling and shear under the fire load combination.

**Available tools:** 3D model

**Results that can be linked:** β_n, d_char,n, d_ef, b_fi, h_fi, k_fi, f_m,d,fi, w_fi, σ_m,fi, k_crit,fi

**Method and assumptions**

- Reduced cross-section: d_ef = β_n t + 7 mm; f_d,fi = k_fi f_k (k_mod,fi = γ_M,fi = 1).

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Strength class | | GL24h | 8 … | |
| Width b | mm | 160 | 60 – … | |
| Depth h | mm | 480 | 100 – … | |
| Span L | m | 6 | 0.5 – … | |
| Exposure | | 3 sides (top protected) | 3 sides (top protected) / 4 sides | |
| Permanent load g_k (incl. self weight) | kN/m | 3 | 0 – … | |
| Variable load q_k | kN/m | 3 | 0 – … | |
| ψ for the fire situation | | 0.3 | 0 – 1 | EN 1991-1-2 4.3.1: ψ2,1 recommended (or ψ1,1 per the National Annex). |
| Required fire resistance | | R30 | R30 / R60 / R90 / R120 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Bending in fire: σ_m,fi ≤ k_crit f_m,d,fi | EN 1995-1-2 4.2.2 | 0.21 |
| Shear in fire: τ_fi ≤ k_fi f_v,k | EN 1995-1-2 4.2.2 | 0.14 |

GEOTECHNICAL

## Berlin wall (soldier pile wall)

Standard: EN 1997-1, EN 1993-1-1, EN 1995-1-1, EN 1998-5

Steel soldier piles with timber lagging, cantilever or with one anchor row: active/passive earth pressure with spatial passive resistance, embedment (Blum / free earth support), pile bending and shear (EC3), lagging (EC5), anchor force, seismic case and indicative deflection. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · automatic design (Pile section, Provided embedment below excavation t)

**Results that can be linked:** H, a, b, z_A, φ'_k, c'_k, γ, q_k, f, b_p, φ'_d, K_ah, K_ph, t_0 …

**Method and assumptions**

- Blum (cantilever) or free earth support (one anchor row); spatial passive resistance below the excavation (f·b ≤ a).

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Excavation depth H | m | 5 | 1 – 15 | |
| Pile spacing a (centre to centre) | m | 2 | 0.5 – 4 | Typical 1.5–3.0 m. |
| Provided embedment below excavation t | m | 3.5 | 0.5 – … | |
| Support system | | Single anchor row | Cantilever / Single anchor row | |
| Anchor depth below top | m | 1.5 | 0.3 – … | |
| Anchor inclination below horizontal | ° | 20 | 0 – 45 | |
| Anchor design resistance R_a,d (per anchor) | kN | 450 | 1 – … | From anchor design / suitability tests (EN 1997-1 §8). |

| | | | | |
|---|---|---|---|---|
| Number of piles shown in 3D | | 6 | 2 – 12 | |
| Pile section | | HEB 240 | 40 … | |
| Steel grade | | S355 | S235 / S275 / S355 | |
| Lagging board thickness | mm | 80 | 30 – 200 | |
| Lagging pressure factor (arching) | | 0.6 | 0.3 – 1 | Arching reduces the pressure on the boards: ~0.6 in competent soils, 1.0 in soft clay or loose sand. |
| Soil φ'k | ° | 32 | 20 – 45 | |
| Soil c'k | kN/m² | 0 | 0 – … | |
| Soil γ | kN/m³ | 19 | 10 – … | |
| Active wall friction δa/φ' | | 0.667 | 0 – 0.667 | |
| Passive wall friction δp/φ' | | 0.333 | 0 – 0.5 | Keep ≤ 0.5: Coulomb overestimates passive resistance. |
| Surcharge behind the wall (variable) | kN/m² | 10 | 0 – … | |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | |
| Include seismic design (EN 1998-5) | | yes | | |
| Design ground acceleration αgR·γI | | 0.16 | 0 – 0.6 | In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex. |
| Soil factor S | | 1.2 | 1 – 1.8 | |
| Displacement factor r | | 1.0 | 1.0 / 1.5 / 2.0 | EN 1998-5 Table 7.1: use 1.0 for anchored / propped walls. |
| ψ2 for surcharge | | 0.3 | 0 – 1 | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| DA2: t_prov ≥ t_req | EN 1997-1 9.7.4 | 0.62 |
| Seismic: t_prov ≥ t_req | EN 1997-1 9.7.4 | 0.8 |
| Pile bending: M_Ed ≤ M_c,Rd | EN 1993-1-1 6.2.5 | 0.42 |
| Pile shear: V_Ed ≤ V_pl,Rd | EN 1993-1-1 6.2.6 | 0.19 |
| Embedment: t_prov ≥ t_req | EN 1997-1 9.7.4 | 0.8 |

| | | |
|---|---|---|
| Anchor: A_d ≤ R_a,d | EN 1997-1 8.5 | 0.39 |
| Lagging: t_prov ≥ t_req | EN 1995-1-1 6.1 | 0.69 |

## Cantilever retaining wall

Standard: EN 1997-1, EN 1998-5, EN 1992-1-1

Reinforced concrete cantilever wall: active earth pressure (Coulomb), surcharge, groundwater, seismic thrust (Mononobe-Okabe), sliding, overturning, bearing (Annex D), and EC2 design of stem, heel and toe. Includes an interactive 3D model.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Toe length, Heel length (behind stem), Base thickness, Stem thickness at base)

**Results that can be linked:** H_stem, t_top, t_bot, t_b, b_toe, b_heel, B, β, h_v, q_k, φ'_d, K_a, φ'_f,d, tan δ_d …

### Method and assumptions

- Coulomb active pressure on the virtual back through the heel; Mononobe-Okabe in the seismic case (±k_v); bearing by Annex D with inclination factors.
- Passive resistance in front is ignored except at a shear key.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Stem height above base | m | 4 | 0.5 – 15 | |
| Stem thickness at top | m | 0.25 | 0.15 – … | |
| Stem thickness at base | m | 0.4 | 0.15 – … | |
| Base thickness | m | 0.5 | 0.2 – … | |
| Toe length | m | 1 | 0 – … | |
| Heel length (behind stem) | m | 2.2 | 0.2 – … | |
| Soil depth over toe | m | 0.5 | 0 – … | Depth of soil in front of the wall above the top of the base. |
| Wall length (3D view) | m | 8 | 1 – … | |
| Stepped stem (thicker lower part) | | no | | Thicken the lower part of the stem where the moments are largest. |
| Height of the step above the base | m | 1.5 | 0.2 – … | |
| Stem thickness below the step | m | 0.55 | 0.2 – … | Above the step the stem tapers from 'thickness at base' to 'thickness at top'. |

| | | | | |
|---|---|---|---|---|
| Shear key under the base | | no | | A downstand under the base that mobilises passive resistance against sliding. |
| Key position (front face, from the toe) | m | 1 | 0 – … | Usually under the stem or just behind it. |
| Key width | m | 0.4 | 0.2 – … | |
| Key depth below the base | m | 0.5 | 0.2 – 2 | |
| Backfill slope β | ° | 0 | 0 – 40 | |
| Backfill φ'k | ° | 30 | 15 – 45 | |
| Backfill γ | kN/m³ | 18 | 10 – … | |
| Backfill γsat | kN/m³ | 20 | 10 – … | |
| Surcharge on backfill (variable) | kN/m² | 10 | 0 – … | e.g. 10 kN/m² for light traffic or storage. |
| Water level above base underside | m | 0 | 0 – … | 0 = drained backfill (no water pressure). |
| Foundation φ'k | ° | 32 | 10 – 45 | |
| Foundation c'k | kN/m² | 0 | 0 – … | |
| Foundation γ | kN/m³ | 19 | 10 – … | |
| Base friction factor k (tan δ = k tan φ') | | 1 | 0.3 – 1 | 1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)). |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | 'National Annex' uses the approach of the annex chosen in the project details. |
| Include seismic design (EN 1998-5) | | yes | | |
| Design ground acceleration αgR·γI | | 0.16 | 0 – 0.6 | In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex. |
| Soil factor S | | 1.2 | 1 – 1.8 | EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4. |
| Displacement factor r | | 2.0 | 1.0 / 1.5 / 2.0 | EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped). |
| ψ2 for surcharge | | 0.3 | 0 – 1 | |
| Concrete class | | C30/37 | C25/30 / C30/37 / C35/45 / C40/50 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Concrete unit weight | kN/m³ | 25 | 20 – … | |

| | | | | |
|---|---|---|---|---|
| Cover — stem (earth face) | mm | 50 | 20 – … | |
| Cover — base | mm | 75 | 25 – … | |
| Stem bars Ø | | 16 | 7 … | |
| Stem bar spacing | mm | 150 | 75 – 400 | |
| Heel top bars Ø | | 16 | 7 … | |
| Heel bar spacing | mm | 150 | 75 – 400 | |
| Toe bottom bars Ø | | 12 | 7 … | |
| Toe bar spacing | mm | 150 | 75 – 400 | |
| Shear key bars Ø | | 12 | 7 … | |
| Shear key bar spacing | mm | 200 | 75 – 400 | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| DA2 sliding: H_d ≤ R_h,d | EN 1997-1 6.5.3 | 0.73 |
| DA2 eccentricity: \|e\| ≤ B/3 | EN 1997-1 6.5.4 | 0.33 |
| DA2 bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.42 |
| Overturning: M_dst,d ≤ M_stb,d | EN 1997-1 Eq. 2.4 | 0.39 |
| Seismic sliding: H_d ≤ R_h,d | EN 1997-1 6.5.3 | 0.92 |
| Seismic eccentricity: \|e\| ≤ B/3 | EN 1997-1 6.5.4 | 0.4 |
| Seismic bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.8 |
| Seismic overturning: M_dst ≤ M_stb | EN 1998-5 7.3.1 | 0.37 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.67 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.2.1.1 | 0.38 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.66 |
| Shear: V_Ed ≤ V_Rd,c | 6.2.2 | 0.51 |
| Bar spacing at max. moment: s ≤ min(2h, 250) | 9.3.1.1(3) | 0.6 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.5 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.2.1.1 | 0.47 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.49 |
| Shear: V_Ed ≤ V_Rd,c | 6.2.2 | 0.44 |
| Bar spacing at max. moment: s ≤ min(2h, 250) | 9.3.1.1(3) | 0.6 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.39 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.2.1.1 | 0.84 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.37 |

| | | |
|---|---|---|
| Shear: V_Ed ≤ V_Rd,c | 6.2.2 | 0.56 |
| Bar spacing at max. moment: s ≤ min(2h, 250) | 9.3.1.1(3) | 0.6 |

## Embedded pile wall with anchors

Standard: EN 1997-1 §9, EN 1992-1-1, EN 1998-5

Contiguous or secant bored pile wall, cantilever or with several anchor rows: elastic–plastic subgrade reaction analysis (displacements, earth pressures, moments, anchor forces), passive mobilisation, RC pile design and anchors. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · automatic design (Embedment below excavation t, Pile diameter D)

**Results that can be linked:** H, t, D, a, EI, k_h, u_max, u_top, P_1,k, P_2,k, M_Ed, V_Ed, E_p,mob, E_p,d …

### Method and assumptions

- Beam on elastic–plastic Winkler springs: at-rest pressure ± k_h u bounded by active and passive; anchors as springs locked off at P_0.
- Factored effects (DA2: characteristic soil, effects × γ_G) or factored strength (DA1-C2, DA3).

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Excavation depth H | m | 7 | 1 – 25 | |
| Embedment below excavation t | m | 5 | 0.5 – … | |
| Pile diameter D | m | 0.6 | 0.3 – 1.5 | |
| Pile spacing a | m | 0.75 | 0.3 – 3 | Contiguous wall: a ≥ D (gap ≤ ~0.15 m in stiff soil); secant: a < D. |
| Anchor rows | table | | | Depth [m], Inclination α [°], Horizontal spacing [m], Lock-off load P_0 [kN], Free length [m], Tendon area A_t [mm²], Design resistance R_a,d [kN] — Leave the table empty for a cantilever wall. Strands Ø15.7: 150 mm² each. |
| Soil φ'k | ° | 32 | 15 – 45 | |
| Soil c'k | kN/m² | 5 | 0 – … | |
| Soil γ | kN/m³ | 19 | 10 – … | |
| Soil γ_sat | kN/m³ | 20 | 10 – … | |
| Groundwater depth behind the wall | m | 20 | 0 – … | In front the water is lowered to the excavation level. |

| | | | | |
|---|---|---|---|---|
| Surcharge behind the wall (variable) | kN/m² | 10 | 0 – … | |
| Modulus of subgrade reaction k_h | MN/m³ | 25 | 2 – 200 | Typical: loose sand 10, medium dense 25, dense 50–100, stiff clay 20–40 MN/m³. |
| Wall friction δ_a/ φ' | | 0.667 | 0 – 0.667 | |
| Wall friction δ_p/ φ' | | 0.5 | 0 – 0.5 | |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | |
| Include seismic design (EN 1998-5) | | no | | |
| Design ground acceleration αgR·γI | | 0.2 | 0 – 0.6 | |
| Soil factor S | | 1.2 | 1 – 1.8 | |
| Concrete class | | C30/37 | C25/30 / C30/37 / C35/45 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Longitudinal bars | | 12 | 6 – 40 | |
| Bar Ø | | 20 | 16 / 20 / 25 / 32 | |
| Spiral Ø | | 10 | 8 / 10 / 12 | |
| Spiral pitch | mm | 150 | 50 – 400 | |
| Cover to bar centre | mm | 85 | 50 – 150 | |
| Bending stiffness factor (cracking) | | 0.7 | 0.3 – 1 | E_cm I × factor; 0.5–0.7 for cracked RC piles. |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| DA2 embedment: E_p,mob ≤ E_p,d | EN 1997-1 9.7.4 | 0.38 |
| DA2 anchor row 1: P_d ≤ R_a,d | EN 1997-1 8.5 | 0.6 |
| DA2 anchor row 2: P_d ≤ R_a,d | EN 1997-1 8.5 | 0.73 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.17 |

| | | |
|---|---|---|
| Shear: V_Ed ≤ max(V_Rd,c, V_Rd,s) | 6.2 | 0.25 |

## Foundation settlement

Standard: EN 1997-1 §6.6, Annex F, Annex H

Immediate and consolidation settlement of a rectangular footing or raft on layered soil: Boussinesq stress distribution, oedometer modulus or C_c/C_r/OCR per layer, influence depth and limits of EN 1997-1 Annex H. 3D stress bulb.

**Available tools:** 3D model

**Results that can be linked:** B, L, D, q, σ'_v0(D), q_net, z_inf, s_c, s

### Method and assumptions

- Boussinesq stresses below the centre (Newmark), sub-layers of 0.25 m down to Δσ = 0.2 σ'_v0; granular by E_oed, clays by C_c/C_r/OCR.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Foundation width B | m | 2.5 | 0.3 – … | |
| Foundation length L | m | 2.5 | 0.3 – … | |
| Founding depth D | m | 1 | 0 – … | |
| Contact pressure (SLS, quasi-permanent) | kN/m² | 180 | 0 – … | Total pressure under the foundation for G + ψ2Q. The net pressure q − γD causes settlement. |
| Rigid foundation (s = 0.8 s_centre) | | yes | | A rigid footing settles uniformly ≈ 0.8 × the flexible centre value. |
| Groundwater depth below ground | m | 4 | 0 – … | |
| Soil layers (from the ground surface down) | table | | | Thickness [m], Model, γ [kN/m³], E_oed (granular) [MPa], C_c (clay), C_r (clay), e_0, OCR |
| Limiting settlement | mm | 50 | 5 – … | EN 1997-1 Annex H: up to 50 mm is often acceptable for normal structures on separate foundations; less for sensitive structures. |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| Settlement: s ≤ s_lim | EN 1997-1 Annex H | 0.34 |

## Gabion wall

Standard: EN 1997-1, EN 1998-5, EN 10223-3

Gravity wall of stacked gabion courses with set-backs: Coulomb and Mononobe-Okabe thrust, sliding, overturning, bearing and the stability of every joint between courses. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · DXF drawing

**Results that can be linked:** H, B, γ_s, n, γ_g, β, h_v, q_k, φ'_d, K_a, φ'_f,d, tan δ_d, H_d, R_h,d ...

### Method and assumptions

- Global stability as a gravity wall (γ_g = γ_s(1 − n)) plus sliding and eccentricity at every joint between courses.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Gabion courses (from the bottom up) | table | | | Width [m], Height [m], Front set-back from the course below [m] — Standard baskets are 0.5 m or 1.0 m high and 1.0–1.5 m wide modules. |
| Embedment of the base below ground in front | m | 0.5 | 0 – ... | |
| Wall length (3D view) | m | 8 | 1 – ... | |
| Backfill slope β | ° | 0 | 0 – 40 | |
| Backfill φ'k | ° | 30 | 15 – 45 | |
| Backfill γ | kN/m³ | 18 | 10 – ... | |
| Backfill γsat | kN/m³ | 20 | 10 – ... | |
| Surcharge on backfill (variable) | kN/m² | 10 | 0 – ... | e.g. 10 kN/m² for light traffic or storage. |
| Water level above base underside | m | 0 | 0 – ... | 0 = drained backfill (no water pressure). |
| Foundation φ'k | ° | 32 | 10 – 45 | |
| Foundation c'k | kN/m² | 0 | 0 – ... | |
| Foundation γ | kN/m³ | 19 | 10 – ... | |
| Base friction factor k (tan δ = k tan φ') | | 1 | 0.3 – 1 | 1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)). |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | 'National Annex' uses the approach of the annex chosen in the project details. |
| Include seismic design (EN 1998-5) | | yes | | |

| | | | | |
|---|---|---|---|---|
| Design ground acceleration αgR·γI | | 0.16 | 0 – 0.6 | In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex. |
| Soil factor S | | 1.2 | 1 – 1.8 | EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4. |
| Displacement factor r | | 2.0 | 1.0 / 1.5 / 2.0 | EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped). |
| ψ2 for surcharge | | 0.3 | 0 – 1 | |
| Unit weight of the stone | kN/m³ | 26 | 18 – 30 | Limestone 24–26, basalt/granite 26–28 kN/m³. |
| Porosity of the fill n | | 0.35 | 0.25 – 0.45 | 0.30–0.40 for hand-placed, well-graded stone. |
| Friction angle between courses | ° | 35 | 20 – 45 | Stone-on-stone contact through the mesh; 30–40°. |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| DA2 sliding: H_d ≤ R_h,d | EN 1997-1 6.5.3 | 0.64 |
| DA2 eccentricity: \|e\| ≤ B/3 | EN 1997-1 6.5.4 | 0.29 |
| DA2 bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.32 |
| Overturning: M_dst,d ≤ M_stb,d | EN 1997-1 Eq. 2.4 | 0.31 |
| Seismic sliding: H_d ≤ R_h,d | EN 1997-1 6.5.3 | 0.8 |
| Seismic eccentricity: \|e\| ≤ B/3 | EN 1997-1 6.5.4 | 0.33 |
| Seismic bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.59 |
| Seismic overturning: M_dst ≤ M_stb | EN 1998-5 7.3.1 | 0.29 |
| Joint 1 sliding: H_Ed ≤ N_fav tan φ_j | EN 1997-1 6.5.3 | 0.42 |
| Joint 1 eccentricity: \|e\| ≤ b_j/3 | EN 1997-1 6.5.4 | 0.03 |
| Joint 2 sliding: H_Ed ≤ N_fav tan φ_j | EN 1997-1 6.5.3 | 0.33 |
| Joint 2 eccentricity: \|e\| ≤ b_j/3 | EN 1997-1 6.5.4 | 0.05 |

## Gravity retaining wall

Standard: EN 1997-1, EN 1998-5, EN 1992-1-1 §12

Mass concrete or masonry gravity wall with optional footing: Coulomb and Mononobe-Okabe thrust, groundwater, sliding, overturning, bearing and plain-concrete section checks. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Width at base of wall body, Footing toe projection, Footing heel projection)

**Results that can be linked:** H, b_top, B_w, f, t_f, B, β, h_v, q_k, φ'_d, K_a, φ'_f,d, tan δ_d, H_d …

### Method and assumptions

- Same stability engine as the cantilever wall; plain-concrete body sections to EN 1992-1-1 §12.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Wall height (above footing) | m | 3 | 0.5 – 10 | |
| Top width | m | 0.5 | 0.2 – … | |
| Width at base of wall body | m | 1.9 | 0.3 – … | |
| Front face batter (horizontal offset) | m | 0.2 | 0 – … | 0 = vertical front face. |
| Footing thickness (0 = no footing) | m | 0.4 | 0 – … | |
| Footing toe projection | m | 0.5 | 0 – … | |
| Footing heel projection | m | 0.3 | 0 – … | |
| Soil depth over toe | m | 0.5 | 0 – … | |
| Wall length (3D view) | m | 8 | 1 – … | |
| Backfill slope β | ° | 0 | 0 – 40 | |
| Backfill φ'k | ° | 30 | 15 – 45 | |
| Backfill γ | kN/m³ | 18 | 10 – … | |
| Backfill γsat | kN/m³ | 20 | 10 – … | |
| Surcharge on backfill (variable) | kN/m² | 10 | 0 – … | e.g. 10 kN/m² for light traffic or storage. |
| Water level above base underside | m | 0 | 0 – … | 0 = drained backfill (no water pressure). |
| Foundation φ'k | ° | 32 | 10 – 45 | |
| Foundation c'k | kN/m² | 0 | 0 – … | |
| Foundation γ | kN/m³ | 19 | 10 – … | |
| Base friction factor k (tan δ = k tan φ') | | 1 | 0.3 – 1 | 1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)). |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | 'National Annex' uses the approach of the annex chosen in the project details. |

| | | | | |
|---|---|---|---|---|
| Include seismic design (EN 1998-5) | | yes | | |
| Design ground acceleration αgR·γI | | 0.16 | 0 – 0.6 | In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex. |
| Soil factor S | | 1.2 | 1 – 1.8 | EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4. |
| Displacement factor r | | 2.0 | 1.0 / 1.5 / 2.0 | EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped). |
| ψ2 for surcharge | | 0.3 | 0 – 1 | |
| Concrete class | | C20/25 | C12/15 / C16/20 / C20/25 / C25/30 / C30/37 | |
| Unit weight of wall | kN/m³ | 24 | 15 – … | 24 plain concrete, 22 stone masonry, 20 gabions (approx.). |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| DA2 sliding: H_d ≤ R_h,d | EN 1997-1 6.5.3 | 0.68 |
| DA2 eccentricity: \|e\| ≤ B/3 | EN 1997-1 6.5.4 | 0.37 |
| DA2 bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.42 |
| Overturning: M_dst,d ≤ M_stb,d | EN 1997-1 Eq. 2.4 | 0.39 |
| Seismic sliding: H_d ≤ R_h,d | EN 1997-1 6.5.3 | 0.83 |
| Seismic eccentricity: \|e\| ≤ B/3 | EN 1997-1 6.5.4 | 0.42 |
| Seismic bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.72 |
| Seismic overturning: M_dst ≤ M_stb | EN 1998-5 7.3.1 | 0.36 |
| Body at base of wall body: N_Ed ≤ N_Rd | 12.6.1 | 0.01 |
| Body at base of wall body: \|e\| ≤ 0.3 b_w | 12.6.5.1 | 0.64 |
| Body at base of wall body: H_Ed ≤ N_Ed tan 35° (joint) | 12.6.3 | 0.46 |
| Body at mid-height: N_Ed ≤ N_Rd | 12.6.1 | 0.01 |
| Body at mid-height: \|e\| ≤ 0.3 b_w | 12.6.5.1 | 0.55 |
| Body at mid-height: H_Ed ≤ N_Ed tan 35° (joint) | 12.6.3 | 0.45 |

## Liquefaction potential

Standard: EN 1998-5 §4.1.4, Annex B

Saturated sandy layers from SPT blow counts: overburden and fines corrections, cyclic resistance with the magnitude factor C_M, seismic shear stress τ_e and the factor of safety against liquefaction for each layer. Interactive 3D soil column.

**Available tools:** 3D model

**Results that can be linked:** a_gR, α, S, αS, C_M, λ

### Method and assumptions

- τ_e = 0.65 α S σ_v0; (N1)60 with C_N and fines correction; CRR of Youd et al. (2001) × C_M (Table B.1); FS ≥ λ of the National Annex.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Reference ground acceleration a_gR | | 0.25 | 0 – 0.6 | In g; zone buttons of the National Annex. |
| Importance class | | II | I / II / III / IV | |
| Soil factor S | | 1.15 | 1 – 1.8 | EN 1998-1 Table 3.2 (ground type C: 1.15). |
| Surface wave magnitude M_s | | 6.5 | 5.5 / 6.0 / 6.5 / 7.0 / 7.5 / 8.0 | Design earthquake magnitude for the site. |
| Groundwater depth | m | 2 | 0 – … | |
| Soil layers (from the surface down) | table | | | Thickness [m], Soil, γ [kN/m³], SPT N_60, Fines content [%] — N_60: SPT blow count corrected to 60 % energy. Clays (FC > 35 % plastic) are not checked. |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| Layer 2 at z = 4.0 m: FS ≥ λ | EN 1998-5 4.1.4(11) | 0.64 |
| Layer 3 at z = 7.5 m: FS ≥ λ | EN 1998-5 4.1.4(11) | 0.79 |
| Layer 4 at z = 12.0 m: FS ≥ λ | EN 1998-5 4.1.4(11) | 0.86 |

## Pad footing

Standard: EN 1997-1, EN 1992-1-1, EN 1998-5

Isolated reinforced concrete footing under a column with biaxial moments: effective-area bearing (Annex D, shape and inclination factors), sliding, eccentricity, bending, beam shear and punching. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Footing length Bx, Footing width By, Footing thickness h)

**Results that can be linked:** B_x, B_y, h, D, c_x, c_y, W_f, W_s, φ'_d, M_x,d, M_y,d, H_d, e_x, e_y …

### Method and assumptions

- Effective area B'×L' for biaxial eccentricity (Annex D); punching with the critical perimeter search between 0 and 2d (6.4.4).

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Footing length Bx | m | 2.4 | 0.5 – … | |
| Footing width By | m | 2.4 | 0.5 – … | |
| Footing thickness h | m | 0.6 | 0.25 – … | |
| Founding depth D (ground to underside) | m | 1.2 | 0.3 – … | |
| Column size cx | m | 0.4 | 0.15 – … | |
| Column size cy | m | 0.4 | 0.15 – … | |
| Axial load — permanent G | kN | 700 | 0 – … | Loads at the top of the footing (column base), downward positive. |
| Axial load — variable Q | kN | 300 | 0 – … | |
| Moment Mx — permanent | kNm | 30 | | Mx causes eccentricity along X (rotation about the Y axis). |
| Moment Mx — variable | kNm | 20 | | |
| Moment My — permanent | kNm | 0 | | My causes eccentricity along Y. |
| Moment My — variable | kNm | 10 | | |
| Horizontal load Hx — permanent | kN | 10 | | |
| Horizontal load Hx — variable | kN | 10 | | |
| Include seismic combination | | yes | | |
| Axial load N_E | kN | 750 | 0 – … | Design values from the seismic analysis (G + ψ2Q + E), capacity design as required by EN 1998-1 4.4.2.6. |
| Moment Mx,E | kNm | 150 | | |
| Moment My,E | kNm | 50 | | |
| Horizontal Hx,E | kN | 80 | | |
| Soil φ'k | ° | 30 | 10 – 45 | |

| | | | | |
|---|---|---|---|---|
| Soil c'k | kN/m² | 5 | 0 – … | |
| Soil γ | kN/m³ | 19 | 10 – … | |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | |
| Concrete class | | C25/30 | C20/25 / C25/30 / C30/37 / C35/45 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Cover (bottom) | mm | 50 | 25 – … | |
| Bars in X Ø | | 14 | 10 / 12 / 14 / 16 / 20 / 25 | |
| Spacing in X | mm | 150 | 75 – 400 | |
| Bars in Y Ø | | 14 | 10 / 12 / 14 / 16 / 20 / 25 | |
| Spacing in Y | mm | 150 | 75 – 400 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| DA2 eccentricity: $(e_x/B_x)^2 + (e_y/B_y)^2 \le 1/9$ | EN 1997-1 6.5.4 | 0 |
| DA2 bearing: $q_{Ed} \le q_{Rd}$ | EN 1997-1 6.5.2 | 0.36 |
| DA2 sliding: $H_d \le R_{h,d}$ | EN 1997-1 6.5.3 | 0.06 |
| Seismic eccentricity: $(e_x/B_x)^2 + (e_y/B_y)^2 \le 1/9$ | EN 1997-1 6.5.4 | 0.08 |
| Seismic bearing: $q_{Ed} \le q_{Rd}$ | EN 1997-1 6.5.2 | 0.42 |
| Seismic sliding: $H_d \le R_{h,d}$ | EN 1997-1 6.5.3 | 0.19 |
| Reinforcement: $A_{s,prov} \ge A_{s,req}$ | 6.1 | 0.6 |
| Minimum reinforcement: $A_{s,prov} \ge A_{s,min}$ | 9.2.1.1 | 0.71 |
| Bending: $M_{Ed} \le M_{Rd}$ | 6.1 | 0.58 |
| Shear: $V_{Ed} \le V_{Rd,c}$ | 6.2.2 | 0.64 |
| Reinforcement: $A_{s,prov} \ge A_{s,req}$ | 6.1 | 0.55 |
| Minimum reinforcement: $A_{s,prov} \ge A_{s,min}$ | 9.2.1.1 | 0.69 |
| Bending: $M_{Ed} \le M_{Rd}$ | 6.1 | 0.54 |
| Shear: $V_{Ed} \le V_{Rd,c}$ | 6.2.2 | 0.61 |
| Punching: $v_{Ed} \le v_{Rd}$ | 6.4.4(2) | 0.58 |
| Punching at column face: $v_{Ed,0} \le v_{Rd,max}$ | 6.4.5(3) | 0.41 |

## Raft foundation on springs

Standard: EN 1997-1 §6, EN 1992-1-1

Raft under a regular column grid on a Winkler subgrade: strip analysis in both directions (settlement, soil pressure, moments), bottom and top reinforcement, punching at the columns and bearing pressure. 3D settlement map.

**Available tools:** 3D model · quantities and cost

**Results that can be linked:** L_x × L_y, t, k_s, q_avg, s_max, q_Ed, m_Ed, μ, μ_lim, ξ, z, A_s,req, A_s,min, A_s,max …

### Method and assumptions

- Strip method: each column line as a beam on Winkler springs carrying the full column loads in both directions (conservative); E_c = 0.5 E_cm.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Columns in x | | 4 | 2 – 10 | |
| Columns in y | | 3 | 2 – 10 | |
| Column spacing in x | m | 5 | 2 – 12 | |
| Column spacing in y | m | 5 | 2 – 12 | |
| Overhang beyond the outer columns | m | 0.8 | 0 – 3 | |
| Raft thickness | mm | 700 | 250 – 2500 | |
| Column size (square) | mm | 450 | 200 – … | |
| Design column load N_Ed (interior) | kN | 2200 | 0 – … | Edge columns take 60 % and corner columns 35 % of this load. |
| Characteristic (SLS) column load | kN | 1550 | 0 – … | |
| Modulus of subgrade reaction k_s | MN/m³ | 15 | 1 – 300 | Soft clay 5–15, medium 15–40, dense sand 40–100 MN/m³. |
| Design bearing resistance q_Rd | kN/m² | 300 | 20 – … | From a bearing capacity calculation (e.g. the strip footing module). |
| Limiting settlement | mm | 50 | 5 – … | |
| Concrete class | | C30/37 | C25/30 / C30/37 / C35/45 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Cover | mm | 50 | 30 – … | |
| Bottom bars Ø (both ways) | | 16 | 12 / 14 / 16 / 20 / 25 | |

| | | | | |
|---|---|---|---|---|
| Bottom bar spacing | mm | 150 | 75 – 300 | |
| Top bars Ø (both ways) | | 16 | 12 / 14 / 16 / 20 / 25 | |
| Top bar spacing | mm | 150 | 75 – 300 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Settlement s_max ≤ s_lim | EN 1997-1 Annex H | 0.1 |
| Bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.35 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.52 |
| Minimum reinforcement | 9.3.1.1 | 0.7 |
| Bending: m_Ed ≤ m_Rd | 6.1 | 0.51 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.44 |
| Minimum reinforcement | 9.3.1.1 | 0.7 |
| Bending: m_Ed ≤ m_Rd | 6.1 | 0.43 |
| Punching: v_Ed ≤ v_Rd | 6.4.4(2) | 0.75 |
| Punching at column face: v_Ed,0 ≤ v_Rd,max | 6.4.5(3) | 0.37 |

## Single pile — axial and lateral

Standard: EN 1997-1 §7, EN 1992-1-1

Bored, CFA or driven pile in layered clay/sand: shaft and base resistance from soil parameters (α-method, β-method, Meyerhof N_q*), partial and correlation/model factors of the National Annex, tension, lateral resistance (Broms) and structural check. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Pile diameter D, Pile length L (below ground))

**Results that can be linked:** D, L, z_w, R_s,cal, q_b, R_b,cal, W_p, ΔW, γ_Rd, R_b,k, R_s,k, F_c,d, R_c,d, M_y …

**Method and assumptions**

- Shaft: α-method (clay) and β-method K tan δ σ'_v (sand); base: 9c_u or Meyerhof N_q* with base efficiency η_b (0.5 bored/CFA, 1.0 driven).
- Characteristic value by model factor (7.6.2.3(8)) or ξ3/ξ4; lateral by Broms (free head, top layer).

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Pile type | | Bored | Bored / CFA / Driven | |
| Pile diameter D | m | 0.8 | 0.2 – 3 | |
| Pile length L (below ground) | m | 18 | 2 – 80 | |
| Concrete class | | C30/37 | C20/25 / C25/30 / C30/37 / C35/45 / C40/50 | |
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Longitudinal bars | | 10 | 6 – 40 | |
| Bar Ø | | 20 | 12 / 14 / 16 / 20 / 25 | |
| Cover to bar centre | mm | 75 | 40 – 150 | Bored piles: cover 75 mm to the links is usual (EN 1536). |
| Axial load — permanent G | kN | 800 | 0 – … | |
| Axial load — variable Q | kN | 250 | 0 – … | |
| Uplift (tension) load — variable | kN | 0 | 0 – … | e.g. wind uplift; checked against shaft resistance in tension. |
| Horizontal load — permanent | kN | 20 | 0 – … | |
| Horizontal load — variable | kN | 30 | 0 – … | |
| Height of the horizontal load above ground e | m | 0 | 0 – … | |
| Groundwater depth below ground | m | 3 | 0 – … | |
| Soil layers (from the ground surface down) | table | | | Thickness [m], Soil, γ [kN/m³], φ' (sand) [°], c_u (clay) [kN/m²] |
| Adhesion factor α (clay) | | 0.5 | 0.2 – 1 | Typical 0.45–0.5 for bored piles in stiff clay; 0.5–1.0 driven. |
| Characteristic resistance | | Alternative method (model factor) | Alternative method (model factor) / Ground | Alternative method: R_k = R_cal / γ_Rd (7.6.2.3(8)). Ground test |

| | | | | |
|---|---|---|---|---|
| | | | test profiles (ξ3, ξ4) | profiles: R_k = min(R_mean/ξ3, R_min/ξ4) (7.6.2.3(5)). |
| Model factor γ_Rd | | 1.4 | 1 – 2 | Cyprus NA 2.19: at least 1.25. About 1.4 when no load tests are made. |
| Number of test profiles n | | 1 | 1 – 10 | |
| Stiff pile cap (ξ / 1.1, 7.6.2.3(7)) | | no | | |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| DA2 compression: F_c,d ≤ R_c,d | 7.6.2.1 | 0.8 |
| DA2 lateral: H_d ≤ H_Rd | EN 1997-1 7.7 | 0.29 |
| Minimum reinforcement A_s ≥ 0.5 % A_c (A_c ≤ 0.5 m²) | EN 1992-1-1 9.8.5 Table 9.6N | 0.76 |
| Axial: N_Ed ≤ N_Rd | 6.1 | 0.14 |
| Bending with axial force: M_Ed ≤ M_Rd(N_Ed) | 6.1 | 0.17 |

## Slope stability (Bishop)

Standard: EN 1997-1 §11, EN 1998-5 §4.1.3

Circular slip surfaces by Bishop's simplified method with automatic search of the critical circle; pore pressure ratio, crest surcharge and seismic pseudo-static action. Interactive 3D model.

**Available tools:** 3D model

**Results that can be linked:** H, β, L_s, φ'_k, c'_k, γ, r_u, q_k, F_k, F', F_d, k_h, k_v, F_d,E

**Method and assumptions**

- Bishop simplified method with automatic search of the critical circle; pseudo-static seismic action (EN 1998-5 4.1.3.3); optional soil nails / anchors adding T·r about the centre.

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Slope height | m | 6 | 1 – 50 | |
| Slope angle | ° | 30 | 5 – 75 | Angle of the slope face to the horizontal (1:√3 ≈ 30°, 1:2 ≈ 26.6°). |

| | | | | |
|---|---|---|---|---|
| Depth to firm stratum below toe | m | 6 | 0 – … | Slip circles are not allowed below this level. |
| Slope length (3D view) | m | 20 | 4 – … | |
| Soil φ'k | ° | 30 | 0 – 45 | |
| Soil c'k | kN/m² | 8 | 0 – … | |
| Soil γ | kN/m³ | 19 | 10 – … | |
| Pore pressure ratio r_u | | 0 | 0 – 0.6 | u = r_u·γ·h. 0 = dry/drained, 0.2–0.3 typical with a high water table. |
| Crest surcharge (variable) | kN/m² | 10 | 0 – … | |
| Distance of surcharge from crest | m | 1 | 0 – … | |
| Surcharge width | m | 5 | 0 – … | |
| Soil nails / anchors (optional) | table | | | Head height above the toe [m], Inclination below horizontal [°], Length [m], Horizontal spacing [m], Design tensile resistance T_d [kN], Design pull-out resistance per metre [kN/m] — Leave empty for an unreinforced slope. Each row adds T = min(T_d, bond × length behind the slip surface) with its lever arm about the circle centre. |
| Include seismic (pseudo-static) | | yes | | |
| Design ground acceleration αgR·γI | | 0.16 | 0 – 0.6 | In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex. |
| Soil factor S | | 1.2 | 1 – 1.8 | |
| Topography factor S_T | | 1 | 1 – 1.4 | EN 1998-5 Annex A: 1.0 for slopes < 15°, up to 1.2–1.4 for high ridges. |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Persistent: E_d ≤ R_d (1 ≤ F_d) | EN 1997-1 Eq. 2.5 | 0.8 |
| Seismic: E_d ≤ R_d (1 ≤ F_d,E) | EN 1998-5 4.1.3.3 | 0.81 |

## Strip footing and tie beams

Standard: EN 1997-1, EN 1992-1-1, EN 1998-5

Continuous footing under a wall (per metre run): bearing with load inclination, sliding, eccentricity, transverse bending and shear. Optional seismic tie beams between footings (EN 1998-5 5.4.1.2).

**Available tools:** 3D model · quantities and cost · DXF drawing · automatic design (Footing width B, Footing thickness h)

**Results that can be linked:** B, h, D, b_w, W_f, W_s, φ'_d, M_d, H_d, e, B', N_q, i_q, q_Rd …

### Method and assumptions

- Strip bearing per metre run with load inclination; tie-beam force ±k α S N (EN 1998-5 5.4.1.2) with the minimum sections of EN 1998-1 5.8.2.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Footing width B | m | 1.4 | 0.4 – … | |
| Footing thickness h | m | 0.45 | 0.2 – … | |
| Founding depth D (ground to underside) | m | 1 | 0.3 – … | |
| Wall thickness b_w | m | 0.25 | 0.1 – … | |
| Line load — permanent G | kN/m | 180 | 0 – … | At the top of the footing (base of the wall), downward positive. |
| Line load — variable Q | kN/m | 60 | 0 – … | |
| Moment — permanent | kNm/m | 5 | | Moment about the footing axis (across the width B). |
| Moment — variable | kNm/m | 5 | | |
| Horizontal load — permanent | kN/m | 0 | | |
| Horizontal load — variable | kN/m | 5 | | |
| Include seismic combination | | no | | |
| Line load N_E | kN/m | 200 | 0 – … | G + ψ2Q + E from the seismic analysis. |
| Moment M_E | kNm/m | 30 | | |
| Horizontal H_E | kN/m | 25 | | |
| Soil φ'k | ° | 30 | 10 – 45 | |
| Soil c'k | kN/m² | 5 | 0 – … | |
| Soil γ | kN/m³ | 19 | 10 – … | |
| Design approach | | National Annex | National Annex / DA1 / DA2 / DA3 | |
| Concrete class | | C25/30 | C20/25 / C25/30 / C30/37 / C35/45 | |

| | | | | |
|---|---|---|---|---|
| Reinforcement fyk | N/mm² | 500 | 400 – 600 | |
| Cover (bottom) | mm | 50 | 25 – … | |
| Transverse bars Ø | | 12 | 10 / 12 / 14 / 16 / 20 / 25 | |
| Spacing of transverse bars | mm | 200 | 75 – 400 | |
| Design tie beams (seismic) | | yes | | Tie beams connect the footings so they move together in an earthquake. |
| Reference ground acceleration a_gR | | 0.25 | 0 – 0.6 | In g. Use the zone buttons. |
| Ground type | | C | B / C / D | Ground type A: tie beams may be omitted (EN 1998-5 5.4.1.2(5)). |
| Importance class | | II | I / II / III / IV | |
| Design axial load of column/wall 1 | kN | 900 | 0 – … | Seismic design situation, the two elements the beam joins. |
| Design axial load of column/wall 2 | kN | 700 | 0 – … | |
| Tie beam width | m | 0.3 | 0.15 – … | |
| Tie beam depth | m | 0.5 | 0.2 – … | |
| Storeys above the basement | | 3 | 1 – 40 | |
| Bars top + bottom (total) | | 8 | 4 – 16 | Half at the top, half at the bottom. |
| Tie beam bars Ø | | 16 | 10 / 12 / 14 / 16 / 20 / 25 | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| DA2 eccentricity: e/B ≤ 1/3 | EN 1997-1 6.5.4 | 0.1 |
| DA2 bearing: q_Ed ≤ q_Rd | EN 1997-1 6.5.2 | 0.56 |
| DA2 sliding: H_d ≤ R_h,d | EN 1997-1 6.5.3 | 0.07 |
| Reinforcement: A_s,prov ≥ A_s,req | 6.1 | 0.5 |
| Minimum reinforcement: A_s,prov ≥ A_s,min | 9.2.1.1 | 0.93 |
| Bending: M_Ed ≤ M_Rd | 6.1 | 0.48 |
| Shear: V_Ed ≤ V_Rd,c | 6.2.2 | 0.33 |
| Tie beam width b ≥ 0.25 m | EN 1998-1 5.8.2(4) | 0.83 |
| Tie beam depth h ≥ 0.4 m (3 storeys) | EN 1998-1 5.8.2(4) | 0.8 |

| | | |
|---|---|---|
| Minimum ratio top and bottom ρ_b ≥ 0.4 % | EN 1998-1 5.8.2(5) | 0.75 |
| Tie beam in tension: N_tie ≤ N_t,Rd | EN 1992-1-1 6.1 | 0.13 |
| Tie beam in compression: N_tie ≤ N_c,Rd | EN 1992-1-1 6.1 | 0.03 |

LOADS

## Load combinations

Standard: EN 1990 Annex A1

Generates all ULS (6.10), seismic (6.12b) and SLS (characteristic, frequent, quasi-permanent) combinations from your actions, with the ψ factors of the National Annex, and finds the governing values.

**Available tools:** 3D model

### Method and assumptions

- All ULS (6.10), seismic (6.12b) and SLS combinations with ψ of the annex; each variable action in turn as leading action.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Actions (characteristic values, one effect e.g. N, M or V) | table | | | Name, Type, Category (imposed), Value |
| Site altitude (for snow ψ) | | 100 | 0 – … | m |
| Unit of the values | | kN | kN / kNm / kN/m / kN/m² | |

## Seismic spectrum and lateral forces

Standard: EN 1998-1

Design response spectrum (Type 1) and the lateral force method: fundamental period, base shear and storey forces for regular buildings.

**Available tools:** 3D model

**Results that can be linked:** a_gR, γ_I, a_g, q, H, T_1, S_d(T_1), λ, W, F_b, F_b / W

### Method and assumptions

- Type 1 design spectrum, T1 = C_t H^¾, λ = 0.85 when applicable, storey forces with a linear mode shape (Eq. 4.11).

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Reference ground acceleration a_gR | | 0.25 | 0 – 0.6 | In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex. |
| Ground type | | B | A / B / C / D / E | A rock, B very dense sand/gravel or stiff clay, C dense to medium-dense, D loose, E soft surface layer on rock. |
| Importance class | | II | I / II / III / IV | II ordinary buildings, III schools/assembly, IV hospitals/civil protection. |
| Behaviour factor q | | 3.9 | 1 – 8 | e.g. RC frame DCM multi-storey: 3.0·αu/α1 = 3.9. |
| Structural system (for T1 = C_t H^¾) | | RC moment frames / eccentric braced steel | Steel moment frames / RC moment frames / eccentric braced steel / All other structures (walls, masonry…) | |
| Fundamental period T1 (0 = estimate) | | 0 | 0 – 4 | s |
| Storeys (height and weight) | table | | | Storey height [m], Seismic weight G + ψE Q [kN] |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| Lateral force method: T_1 ≤ min(4T_C, 2.0 s) | 4.3.3.2.1(2) | 0.24 |

## Wind and snow loads at the site

Standard: EN 1991-1-4, EN 1991-1-3

Peak velocity pressure q_p(z), wall pressures and roof snow load. With the Cyprus National Annex v_b,0 and s_k are taken automatically from the project site on the official maps.

**Available tools:** 3D model

**Results that can be linked:** v_b,0, v_b, z_0, k_r, c_r(z), v_m(z), I_v(z), q_p(z), c_e(z), h/d, c_pe,10 (D), c_pe,10 (E), w_D, w_E …

### Method and assumptions

- q_p(z) by Eq. 4.8 with c_dir = c_season = 1; walls by Table 7.1; roof snow μ_1 C_e C_t s_k.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Use the project site (maps of the National Annex) | | yes | | Pick the site in the project details (📍 Map). Otherwise give v_b,0 and s_k. |
| Fundamental basic wind velocity v_b,0 | | 24 | 10 – 60 | m/s — used when no site is selected or for EN values. |
| Site altitude above sea level | | 100 | 0 – 3000 | m |
| Ground snow load s_k (manual) | kN/m² | 0.5 | 0 – … | Used only when the National Annex gives no snow map/formula. |
| Terrain category | | III | 0 / I / II / III / IV | 0 sea, I lakes/flat, II open country, III suburbs/villages, IV cities (≥15 % covered by buildings > 15 m). |
| Orography factor c_o | | 1 | 1 – 2 | 1.0 unless the site is on a hill or cliff (Annex A.3). |
| Reference height z (building height) | m | 9 | 1 – 200 | |
| Building depth in the wind direction d | m | 12 | 1 – … | |
| Roof pitch α | ° | 0 | 0 – 80 | |
| Exposure (snow) | | Normal | Windswept / Normal / Sheltered | |

## MASONRY

### Masonry wall

Standard: EN 1996-1-1

Load-bearing unreinforced masonry wall: f_k from units and mortar, γ_M by category and class, effective height and thickness, slenderness, reduction factors at top/bottom/mid-height (Annex G) and in-plane shear of a shear wall. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · automatic design (Wall thickness t)

**Results that can be linked:** K, f_k, γ_M, f_d, ρ_n, h_ef, t_ef, h_ef / t_ef, N_m,d, e_init, e_i,top, e_i,bot, Φ_top, e_k …

### Method and assumptions

- Vertical resistance with the reduction factors Φ at top, bottom and mid-height (Annex G, E = K_E f_k); in-plane shear by Eq. 6.13.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Masonry unit | | Clay | Clay / Calcium silicate / Aggregate concrete / Autoclaved aerated concrete / Manufactured stone / Dimensioned natural stone | |
| Unit group | | Group 2 | Group 1 / Group 2 / Group 3 / Group 4 | Group 1 solid (voids ≤ 25 %), 2 vertically perforated ≤ 55 %, 3 ≤ 70 %, 4 horizontally perforated. |
| Normalised unit strength f_b | N/mm² | 10 | 2 – 60 | |
| Mortar class | | M10 | M2.5 / M5 / M10 / M15 / M20 | |
| Units and mortar | | Category I units, designed mortar | Category I units, designed mortar / Category I units, prescribed mortar / Category II units, any mortar | |
| Execution control class | | 2 | 1 / 2 / 3 / 4 / 5 | 1 = best supervision … 5 = least. |
| Final creep coefficient φ∞ | | 1 | 0.3 – 3 | Clay 0.5–1.5, concrete/calcium silicate 1.0–2.0 (Table 6 CYS). |
| Wall thickness t | mm | 250 | 90 – 600 | |

| | | | | |
|---|---|---|---|---|
| Clear storey height h | m | 3 | 1 – 8 | |
| Wall length l_w | m | 4 | 0.3 – … | |
| Restraint of the wall | | Top and bottom (RC floors) | Top and bottom (RC floors) / Top and bottom (timber floors) / Top, bottom and one vertical edge / Top, bottom and both vertical edges | |
| Design vertical load at the top N_Ed | kN/m | 180 | 0 – … | |
| Eccentricity of the floor load at the top | mm | 15 | 0 – … | From the floor bearing (e.g. t/6 for a slab bearing on part of the wall). |
| In-plane shear on the wall V_Ed | kN | 60 | 0 – … | 0 = not a shear wall. |
| Unit weight of masonry | kN/m³ | 18 | 5 – 25 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Slenderness: h_ef/t_ef ≤ 27 | 5.5.1.4 | 0.33 |
| Vertical load at top: N_Ed ≤ Φ t f_d | Eq. 6.2 | 0.32 |
| Vertical load at mid-height: N_Ed ≤ Φ t f_d | Eq. 6.2 | 0.34 |
| Vertical load at bottom: N_Ed ≤ Φ t f_d | Eq. 6.2 | 0.33 |
| In-plane shear: V_Ed ≤ V_Rd | 6.2 | 0.17 |

STEEL

## Bolted connection

Standard: EN 1993-1-8 §3

Bolt group in single or double shear with eccentric shear and tension: spacing rules, bolt shear, bearing, tension, punching, combined shear and tension, long joints, block tearing and the net

section of the plate. Interactive 3D model with bolt forces.

**Available tools:** 3D model

**Results that can be linked:** d_0, f_ub, f_u, V_Ed / n, F_v,Ed, F_v,Rd, F_b,Rd, b, N_pl,Rd, N_u,Rd, V_eff,1,Rd

### Method and assumptions

- Elastic distribution of eccentric shear in the bolt group; bearing checked for the governing bolt with the end/inner values of Table 3.4.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Bolt class | | 8.8 | 4.6 / 5.6 / 8.8 / 10.9 | |
| Bolt diameter | | M20 | 8 … | |
| Threads in the shear plane | | yes | | |
| Shear planes | | 1 | 1 / 2 | 2 = double cover plates (splice). |
| Bolt rows in the force direction n1 | | 3 | 1 – 12 | |
| Bolt columns across n2 | | 2 | 1 – 8 | |
| Pitch p1 (in force direction) | mm | 70 | 20 – … | |
| Pitch p2 (across) | mm | 70 | 20 – … | |
| End distance e1 | mm | 50 | 10 – … | |
| Edge distance e2 | mm | 40 | 10 – … | |
| Thinnest plate thickness t | mm | 12 | 4 – 60 | |
| Plate steel grade | | S275 | S235 / S275 / S355 | |
| Shear force V_Ed (along the force direction) | kN | 300 | 0 – … | |
| Eccentricity of V from the bolt-group centroid | mm | 0 | 0 – … | Causes a torsional moment on the group (elastic method). |
| Tension per bolt F_t,Ed | kN | 0 | 0 – … | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| e_1 ≥ 1.2 d_0 | Table 3.3 | 0.53 |
| e_2 ≥ 1.2 d_0 | Table 3.3 | 0.66 |
| p_1 ≥ 2.2 d_0 | Table 3.3 | 0.69 |
| p_2 ≥ 2.4 d_0 | Table 3.3 | 0.75 |
| p_1, p_2 ≤ min(14t, 200) | Table 3.3 | 0.42 |

| | | |
|---|---|---|
| Bolt shear: F_v,Ed ≤ F_v,Rd | Table 3.4 | 0.53 |
| Bearing: F_v,Ed ≤ F_b,Rd | Table 3.4 | 0.32 |
| Group: V_Ed ≤ n min(F_v,Rd, F_b,Rd) | 3.7 | 0.53 |
| Plate tension: V_Ed ≤ min(N_pl,Rd, N_u,Rd) | EN 1993-1-1 6.2.3 | 0.76 |
| Block tearing: V_Ed ≤ V_eff,1,Rd | 3.10.2 | 0.42 |

## Column base plate

Standard: EN 1993-1-8 §6.2.5–6.2.8, EN 1992-1-1 §6.7

Base plate of an I/H column on a concrete foundation under axial force, strong-axis moment and shear: bearing strength f_jd, effective width c, compression T-stubs, tension T-stubs with anchor bolts, moment resistance and shear by friction and bolts. Interactive 3D model.

**Available tools:** 3D model · quantities and cost · automatic design (Plate thickness t_p)

**Results that can be linked:** k_j, f_jd, c, F_C,Rd, N_j,Rd, m_x, l_eff,1, M_pl,1,Rd, F_T,1-2,Rd, F_T,3,Rd, F_T,Rd, z, F_t,Ed, F_c,Ed ...

### Method and assumptions

- Component method: compression T-stubs on the foundation, tension T-stub with anchor bolts (no prying), M–N from statics about the compressed flange.
- Anchorage of the bolts in concrete (EN 1992-4) and welds are not checked.

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Column section | | HEB 240 | 40 ... | |
| Axial compression N_Ed | kN | 600 | 0 – ... | |
| Moment M_y,Ed | kNm | 100 | 0 – ... | |
| Shear V_Ed | kN | 40 | 0 – ... | |
| Plate width b_p (along the flanges) | mm | 400 | 150 – ... | |
| Plate length l_p (along the web) | mm | 500 | 150 – ... | |
| Plate thickness t_p | mm | 25 | 8 – 80 | |
| Plate steel grade | | S275 | S235 / S275 / S355 | |
| Anchor bolt class | | 8.8 | 4.6 / 5.6 / 8.8 / 10.9 | |
| Anchor bolt diameter | | M24 | 7 ... | |
| Bolts per side (in each tension row) | | 2 | 1 – 4 | |
| Bolt distance from plate end e_x | mm | 50 | 25 – ... | |
| Bolt gauge across w | mm | 240 | 60 – ... | |
| Foundation concrete | | C25/30 | C20/25 / C25/30 / C30/37 / C35/45 | |

| | | | | |
|---|---|---|---|---|
| Foundation width | m | 1.2 | 0.3 – … | |
| Foundation length | m | 1.2 | 0.3 – … | |
| Flange weld throat a | mm | 8 | 3 – 20 | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Pure compression: N_Ed ≤ N_j,Rd | 6.2.8.2 | 0.28 |
| Tension side: F_t,Ed ≤ F_T,Rd | Table 6.7 | 0.44 |
| Compression side: F_c,Ed ≤ F_C,Rd | Table 6.7 | 0.69 |
| Shear: V_Ed ≤ F_f,Rd + F_vb,Rd | 6.2.2(8) | 0.12 |

## Fillet weld group

Standard: EN 1993-1-8 §4.5

Rectangular group of fillet welds under axial force, shear and bending: throat stresses σ⊥, τ⊥, τ∥ by the directional method (Eq. 4.1) and the simplified method, with the weld detailing rules. Interactive 3D model.

**Available tools:** 3D model · quantities and cost

**Results that can be linked:** A_w, I_w, f_u, β_w, σ_w, σ⊥ = τ⊥, τ∥, σ_eq, f_vw,d, F_w,Ed / a, utilisation

**Method and assumptions**

- Elastic stresses on the throat area unfolded into the joint plane; σ⊥ = τ⊥ = σ_w/√2, τ∥ from the shear on the vertical welds; Eq. 4.1 and the simplified method 4.5.3.3.

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Width of the weld group b (horizontal welds) | mm | 200 | 20 – … | |
| Height of the weld group h (vertical welds) | mm | 300 | 20 – … | |
| Weld along the top (horizontal) | | yes | | |
| Weld along the bottom (horizontal) | | yes | | |
| Welds along both sides (vertical) | | yes | | |
| Throat thickness a | mm | 5 | 3 – 20 | a ≥ 3 mm (4.5.2(2)); leg length ≈ 1.41 a. |
| Steel grade (weaker part) | | S275 | S235 / S275 / S355 | |

| | | | | |
|---|---|---|---|---|
| Thickness of the thinner part | mm | 12 | 3 – … | |
| Axial force N_Ed (tension +) | kN | 150 | | |
| Shear force V_Ed (vertical) | kN | 120 | 0 – … | |
| Moment M_Ed (about the horizontal axis) | kNm | 20 | | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Throat a ≥ 3 mm | 4.5.2(2) | 0.6 |
| Throat a ≤ 0.7 t (practical maximum) | good practice | 0.6 |
| Weld length b ≥ max(30 mm, 6a) | 4.5.4(1) | 0.15 |
| Weld length h ≥ max(30 mm, 6a) | 4.5.4(1) | 0.1 |
| Weld: $\sqrt{(\sigma_\perp^2 + 3(\tau_\perp^2 + \tau_\parallel^2))} \le f_u/(\beta_w \gamma_{M2})$ | Eq. 4.1 | 0.31 |
| Weld: $\sigma_\perp \le 0.9 f_u / \gamma_{M2}$ | Eq. 4.1 | 0.17 |

## Steel beam design

Standard: EN 1993-1-1:2005

Simply supported rolled I-beam: classification, bending, shear, bending-shear interaction, lateral-torsional buckling and deflection.

**Available tools:** 3D model · quantities and cost · automatic design (Section)

**Results that can be linked:** f_y, ε, E, G, c_f/t_f, c_w/t_w, W_y, g_sw, g_k, q_k, w_Ed, M_Ed, V_Ed, M_c,Rd …

**Method and assumptions**

- Simply supported rolled I-beam; LTB with M_cr (NCCI SN003) and 6.3.2.3.

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Section | | IPE 300 | 40 … | |
| Steel grade | | S275 | S235 / S275 / S355 / S460 | |
| Span | m | 6 | 0.1 – … | |
| LTB effective length (0 = unrestrained span) | m | 2 | 0 – … | Distance between lateral restraints to the compression flange. |
| Permanent UDL (excl. self weight) | kN/m | 8 | 0 – … | |
| Variable UDL | kN/m | 10 | 0 – … | |

| | | | | |
|---|---|---|---|---|
| Permanent point load at midspan | kN | 0 | 0 – … | |
| Variable point load at midspan | kN | 0 | 0 – … | |
| γG | | 1.35 | 0 – … | |
| γQ | | 1.5 | 0 – … | |
| γM0 | | 1 | 0.5 – … | |
| γM1 | | 1 | 0.5 – … | |
| Variable load deflection limit L/… | | 360 | 1 – … | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Bending: M_Ed ≤ M_c,Rd | 6.2.5(1) | 0.69 |
| Shear: V_Ed ≤ V_pl,Rd | 6.2.6(1) | 0.19 |
| Shear buckling: h_w/t_w ≤ 72ε/η | 6.2.6(6) | 0.59 |
| LTB: M_Ed ≤ M_b,Rd | 6.3.2.1(1) | 0.73 |
| Deflection: δ_q ≤ δ_lim | EN 1990 A1.4 | 0.58 |

## Steel column

Standard: EN 1993-1-1 §6.3, Annex B

Rolled I/H column under axial force and biaxial end moments: classification, cross-section resistance, flexural buckling about both axes, lateral-torsional buckling and the interaction of Eq. 6.61/6.62 (Annex B). Interactive 3D model with buckled shape.

**Available tools:** 3D model · quantities and cost · automatic design (Section)

**Results that can be linked:** f_y, ε, c/t (flange), c/t (web), N_c,Rd, M_y,Rd, M_z,Rd, V_pl,z,Rd, L_cr,y, λ̄_y, χ_y, N_b,y,Rd, L_cr,z, λ̄_z …

**Method and assumptions**

- Buckling curves of Table 6.2; interaction factors of Annex B (Method 2).
- Class 4 webs by the effective width of EN 1993-1-5.

**Inputs**

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Section | | HEB 240 | 40 … | |
| Steel grade | | S355 | S235 / S275 / S355 / S460 | |

| | | | | |
|---|---|---|---|---|
| Column height L | m | 4 | 0.5 – 30 | |
| Buckling length factor about y-y | | 1 | 0.5 – 3 | 1.0 pinned–pinned, 0.7 fixed–pinned, 0.5 fixed–fixed, 2.0 cantilever; sway frames > 1.0. |
| Buckling length factor about z-z | | 1 | 0.3 – 3 | Bracing at mid-height about the weak axis → 0.5. |
| LTB length factor | | 1 | 0.3 – 3 | |
| Axial compression N_Ed | kN | 1200 | 0 – … | |
| M_y,Ed at top | kNm | 80 | | Strong-axis moment; same sign at both ends = single curvature. |
| M_y,Ed at bottom | kNm | -40 | | |
| M_z,Ed at top | kNm | 5 | | |
| M_z,Ed at bottom | kNm | 0 | | |
| Shear V_z,Ed | kN | 30 | 0 – … | |

**Checks performed (worked example)**

| Check | Clause | U (example) |
|---|---|---|
| Cross-section: N/N_Rd + M_y/M_y,Rd + M_z/M_z,Rd ≤ 1 | Eq. 6.2 | 0.56 |
| Shear: V_Ed ≤ V_pl,Rd | 6.2.6 | 0.04 |
| Buckling about y-y: N_Ed ≤ N_b,Rd | 6.3.1 | 0.36 |
| Buckling about z-z: N_Ed ≤ N_b,Rd | 6.3.1 | 0.51 |
| LTB: M_y,Ed ≤ M_b,Rd | 6.3.2.1 | 0.21 |
| Eq. 6.61: N/(χ_y N_Rk/γ_M1) + k_yy M_y/(χ_LT M_y,Rk/γ_M1) + k_yz M_z/M_z,Rd ≤ 1 | 6.3.3(4) | 0.47 |
| Eq. 6.62: N/(χ_z N_Rk/γ_M1) + k_zy M_y/(χ_LT M_y,Rk/γ_M1) + k_zz M_z/M_z,Rd ≤ 1 | 6.3.3(4) | 0.69 |

## TIMBER

### Timber beam

Standard: EN 1995-1-1, EN 338, EN 14080

Simply supported solid timber or glulam beam: k_mod by service class and load duration, size factor, bending with lateral-torsional buckling (k_crit), shear (k_cr), bearing (k_c,90) and instantaneous/final deflection with creep (k_def). Interactive 3D model.

**Available tools:** 3D model · quantities and cost · automatic design (Depth h, Width b)

**Results that can be linked:** γ_M, g_sw, M_Ed, V_Ed, k_h, f_m,d, σ_m,d, l_ef, σ_m,crit, λ_rel,m, k_crit, τ_d, f_v,d, k_c,90 …

### Method and assumptions

- Governing combination by the largest w_d / k_mod; deflection includes shear deformation and creep (k_def).

### Inputs

| Input | Unit | Example | Range | Notes |
|---|---|---|---|---|
| Strength class | | GL24h | 8 … | |
| Width b | mm | 140 | 38 – 400 | |
| Depth h | mm | 400 | 75 – 2000 | |
| Span L | m | 6 | 0.5 – 40 | |
| Lateral restraint spacing (0 = span) | m | 0 | 0 – … | Distance between restraints of the compression edge (e.g. purlins, floor joists). |
| Load on the compression (top) edge | | yes | | Destabilising: l_ef + 2h (Table 6.1). |
| Support length | mm | 120 | 40 – … | |
| Service class | | 1 | 1 / 2 / 3 | 1 heated interior, 2 covered unheated, 3 exposed outdoors. |
| Permanent load g_k (excl. self weight) | kN/m | 2 | 0 – … | |
| Variable load q_k | kN/m | 3 | 0 – … | |
| Duration of the variable load | | Medium-term | Permanent / Long-term / Medium-term / Short-term / Instantaneous | Imposed floor load: medium; snow: short (Cyprus); wind: short/instant. |
| ψ2 of the variable load | | 0.3 | 0 – 1 | |
| Limit w_inst (variable) = L / … | | 300 | 100 – … | |
| Limit w_net,fin = L / … | | 250 | 100 – … | |

### Checks performed (worked example)

| Check | Clause | U (example) |
|---|---|---|
| Bending: σ_m,d ≤ f_m,d | Eq. 6.11 | 0.57 |
| LTB: σ_m,d ≤ k_crit f_m,d | Eq. 6.33 | 0.57 |
| Shear: τ_d ≤ f_v,d | Eq. 6.13 | 0.4 |
| Bearing: σ_c,90,d ≤ k_c,90 f_c,90,d | Eq. 6.3 | 0.38 |
| w_inst,Q ≤ L/300 | Table 7.2 | 0.32 |

| | | |
|---|---|---|
| w_net,fin ≤ L/250 | Table 7.2 | 0.63 |